Affordances for Harm

How Offenders Misuse Platform Capabilities to Exploit Children, and Where to Intervene

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Corpus	7,426 cases · 80,000+ features · 56 sources
Coverage	61 task forces · 3,500+ law enforcement agencies
PACER	30 federal prosecution records · 27 federal indictments · 8 Statements of Offense
Timespan	2002–2026
Platforms	30+ analyzed

Abstract

For years, the history of the internet has been championed through the lens of platforms, growth, and innovation. Far less often do we systematically examine the harms that have scaled alongside that same infrastructure.

From large-scale content distribution and global connectivity to encrypted messaging and AI-generated imagery, the capabilities that make technology worth using have also made it exploitable. The same infrastructure that connected billions of people quietly rewired the economics of child exploitation, not because these technologies were designed to cause harm, but because offenders are adaptive: some pursue children deliberately, others are capitalizing on access and opportunity the internet has made possible for the first time.

This paper draws on 7,426 curated Internet Crimes Against Children case records spanning 2002 to 2026 and asks: which platform capabilities offenders exploit most consistently, how technology is weaponized within specific offense subtypes—grooming, sextortion, production of CSAM, coordinated criminal networks—and where disruption is most realistic. Across 30+ platforms and 61 task forces, the same capability types surface again and again—anonymity, disappearing content, file distribution, contact discovery, trust-building at speed—regardless of which platform is in the headlines. The technology changes. The exploitation mechanics do not.

Contents

1. Exploitation Is a Human Choice

The capabilities that make the internet valuable to hundreds of millions of legitimate users are the same capabilities offenders have learned to turn against children. Every case in this corpus begins with the same decision: a person chose to exploit a child.

Across the twenty-four years and 7,426 enforcement records this corpus spans, what technology changed was not that decision but its operational scope.

Child sexual exploitation is not a problem the internet invented. It is among the oldest documented harms to children, and the children it targets have always been vulnerable for the same reasons: age, developmental stage, and dependency on adults. What the internet changed is the capability available to the person who decides to act on that vulnerability. The offender in 1990 operated inside constraints that no longer exist in the same form. Physical proximity was required to make contact. Geography bounded victim access. Material was difficult to produce, store, and distribute. Detection risk was comparatively high at every step. In 2026, none of those constraints hold. A device in a pocket provides global reach. Anonymity is default on the most-used platforms in the world. Content can be produced, encrypted, distributed, and erased in seconds. The barriers did not fall because technology made people worse. They fell because technology raised the capability ceiling for everyone, including those who use it to harm.

This is the amplification problem, and it is distinct from causation. No platform built a feature to enable child exploitation. The features exist for hundreds of millions of legitimate users, and bad actors have learned to use the same capabilities that make the technology valuable. That is not a design failure unique to any company; it is the reality of the modern digital society. It is also why the question *which platform did this?* is less useful than the question this paper asks: *which capabilities are consistently reached for, and why do they appear across every platform in the enforcement record?*

Offenders in this corpus range from individuals who systematically identified and groomed multiple victims across years to those who acted on an available opportunity technology placed in front of them. Both types are in the enforcement record. Both made a choice. Understanding how malicious users transform platform affordances into misuse surfaces is the prerequisite for designing against them.

The landscape of exploitation continues to evolve alongside technology. The dominant image—a predator posing as a peer, deceiving a child into a meeting that ends in assault—is one offense type in the enforcement record. It is not the only one. Sextortion, in which offenders coerce victims through threats to expose images, emerged as a distinct and rapidly growing typology only in the last decade, made possible by the same digital infrastructure that connects everyone online. Coordinated networks of offenders operating across gaming platforms represent another. Each capability introduced has, in turn, produced offense patterns that did not previously exist at scale. Offending profiles and capabilities are not static. They expand with technology.

The harm in these cases is not abstract. Livingstone and Smith [9] document the inseparability of digital and physical risk for children: what happens on a platform shapes what happens offline, and the boundary between them is not meaningful to the child living inside it. Contact made in a chat room migrates. Material produced in one context circulates in others long after the offense ends. The platforms offenders weaponized did not manufacture predatory intent; they shortened the path from intent to harm.

This paper asks three sequential questions about modern exploitation. Which capabilities do offenders consistently reach for, across offense types, platforms, and years? How do those affordances map to harm vectors across the technological eras this corpus spans? Where does intervention become realistic? The affordance framework developed in §2 provides the analytical language: a way to ask not which platform was involved but which capability was used, and what that capability made possible. The answer turns out to be stable across three decades of platform evolution. Anonymity. Ephemerality. Distribution infrastructure. Contact discovery. Trust architecture at speed. These capabilities appear in the enforcement record because they are structurally useful: they translate platform properties into offense mechanics.

Exploitation is a human choice. The technology does not make that choice. But it shapes, in ways this paper measures at scale for the first time, the harm signatures of modern offenders.

2. The Infrastructure of Harm

2.1 Prior Work: CSEA, Platforms, and the Enforcement Ecosystem

Online child exploitation is one of the most extensively documented crime categories in the federal enforcement ecosystem. It is also one of the least studied at the case level. Understanding why requires understanding what the existing infrastructure does, and challenges the system faces.

How ICAC Enforcement Works. The primary law enforcement response in the United States is organized through the Internet Crimes Against Children Task Force Program: 61 federally coordinated task forces representing over 5,400 federal, state, and local agencies, established in 1998 alongside the NCMEC CyberTipline and expanded continuously since. In fiscal year 2024, ICAC task forces conducted approximately 203,467 investigations and arrested more than 12,600 suspected offenders [14]. Investigations are classified as proactive—investigators initiating contact in undercover capacity—or reactive, triggered by a tip or complaint. The FBI's Violent Crimes Against Children program and Homeland Security Investigations operate in parallel for multi-jurisdictional cases.

The statutory framework has three main layers. The PROTECT Our Children Act of 2008 codified the ICAC program and established mandatory reporting requirements. 18 U.S.C. § 2258A requires electronic service providers to report apparent CSAM to the CyberTipline. The REPORT Act of 2024 extended those obligations to online enticement and child sex trafficking for the first time. In 2024, NCMEC received approximately 20.5 million CyberTipline reports, equivalent to 29.2 million separate incidents, from platforms operating globally [11]. Against that volume, 12,600 arrests represents successful case resolution for a fraction of what is reported. Modern reporting pipelines surface suspected exploitation faster than investigators can clear it, and triage quality determines which offenders are caught and which victims get reached.

The Response Infrastructure. The technology ecosystem for child protection operates on three planes. The first is content detection. PhotoDNA, developed by Microsoft in 2009, generates perceptual hashes of images and compares them against a database of known CSAM, detecting matches even after modification. It is deployed by Google, Meta, Twitter, Reddit, Discord, and dozens of other platforms. Google's CSAI Match performs the equivalent function for video. Meta's open-sourced PDQ and TMK+PDQF algorithms extend perceptual hashing to smaller platforms that cannot build detection infrastructure independently. Thorn's Safer platform takes a complementary approach: perceptual hashing against a database of over 82 million verified CSAM hashes combined with an AI-based classifier designed to identify novel material that has never been hashed, with cross-platform hash-sharing that propagates newly identified CSAM across the detection ecosystem without delay [21].

According to the Tech Coalition's 2023 annual survey, 89% of member companies deploy at least one image hash-matcher; 59% use video hash-matching [20]. This infrastructure is the backbone of CSAM detection at scale, and its effectiveness for known material is well-established [18]. It also has four structural limits that become more significant as the threat landscape evolves. Hash-matching requires a prior identification: new material, including AI-augmented content, has no hash to match. Detection only occurs on cooperating platforms. Encrypted messaging services, smaller platforms, and dark web networks are outside the detection envelope. Hash-matching is also increasingly vulnerable to adversarial attack: recent research has demonstrated that gradient-based image modifications can defeat PhotoDNA detection while leaving content visually intact [2]. And hash-matching answers exactly one question: is this image known illegal material? It cannot answer any of the investigatively important questions — which platform capability made this offense possible, who created this material, how does this case relate to the hundreds investigated last year across the same platform.

*The Behavioral Research Base.*The second major body of work is empirical. The Crimes Against Children Research Center at the University of New Hampshire produced the most systematic research on online CSEA offending. Wolak et al. [24] documented the dynamics of internet-initiated sex crimes through a national law enforcement survey: who offends, how online relationships develop, how offenses progress from digital contact to physical harm. Wolak et al. [25] established that most internet-initiated sex offenses do not fit the predator-deception model that dominates prevention messaging. Most offenders did not disguise their age or intentions but cultivated relationships over time, exploiting adolescent developmental vulnerability rather than deceiving victims about who they were. Wolak and Finkelhor [23] and Wolak et al. [26] characterized sextortion as an emerging and distinct offense typology with its own dynamics and victim population. Kloess et al. [7] provided a comprehensive framework for the process of online sexual grooming across internet communication platforms: prevalence, offense stages, offender characteristics, and legal responses across jurisdictions. Kloess et al. [8] mapped offense processes specifically through internet communication platforms, tracing how offenders move victims from initial contact through escalation. Livingstone and Smith [9] documented the intertwining of digital and physical risk for children at national scale. Mitchell et al. [10] characterized the law enforcement response to internet-facilitated commercial sexual exploitation using a nationally representative sample of agencies.

This body of work established the behavioral landscape of online CSEA. The affordance lens itself has been applied to CSEA conceptually, by psychologists modeling technology-mediated abuse [16], and by criminologists theorizing the mechanics through which technological design "invites" technology-facilitated violence [27]. What no prior work has done is connect exploitation patterns to platform capabilities via an affordance-level abstraction, operationalized against enforcement case records at scale.

*Operational Forensics.*The third body of work is forensics. After a device is seized, investigators use enterprise forensics platforms to examine it: Nuix for large-scale e-discovery and timeline reconstruction, Magnet Axiom for mobile and cloud forensics, Cellebrite UFED for physical and logical extraction. The workflow is precise — physical images of storage media, file carving to recover deleted content and fragments, parsing of application databases and encrypted chat logs, artifact recovery from unallocated space, timeline reconstruction from access metadata and system logs. These platforms are powerful, expensive, and infrastructure-heavy. They ask what is on this phone. Not what this case shares with five hundred cases investigated last year across the same platform.

All three layers of this infrastructure carry an acute human cost. Investigators in child exploitation units routinely encounter material that has no equivalent in other areas of law enforcement. Digital forensics analysts do not merely view this material; they analyze it, build timelines, and construct the evidentiary record that prosecution requires, with consequences including emotional numbing, hypervigilance, intrusive ideation, and accelerated burnout [19]. The volume of evidence has grown as the threat landscape has expanded: a single device seizure now routinely contains hundreds of thousands of images and terabytes of cloud-synced material. The cognitive and psychological load scales accordingly. Triage tools that reduce manual exposure are not only an efficiency argument. They are a wellbeing argument.

The harm for victims in these cases does not end with the offense. Material produced circulates long after investigations close. Survivors navigate criminal proceedings, civil systems, and the ongoing reality that digital content does not disappear. Investigators carry the cases they have worked. Families carry what those cases revealed about someone they trusted. These are not incidental facts about a difficult field. They are part of why the harms this paper addresses matter. Every unworked CyberTip is a case that may go uninvestigated. Every missed cross-case pattern and unreported platform capability is an offense trajectory that could have been identified earlier. The infrastructure described here — detection, enforcement, behavioral research, operational forensics — is staffed by capable, motivated professionals working under immense constraints of scale, an evolving threat landscape, and the sustained weight of radioactive material. The following section introduces the ontological and affordance-based framework that makes cross-case, platform-level analysis possible at corpus scale — without the direct exposure that makes engagement in this field so difficult to sustain.

2.2 What Technology Makes Possible

"Platform" is the wrong unit of analysis for this problem.

Platforms emerge, scale, and disappear. Myspace is gone. Omegle was shut down. Discord did not exist fifteen years ago. If the analytical unit is the platform, every new platform generation requires research from scratch. The offense mechanics do not restart with each platform generation. The capabilities that appear consistently in the enforcement record — anonymity, ephemerality, unmonitored communication, contact discovery, distribution infrastructure — appear on every platform, implemented differently but functioning in similar manners. The unit that holds across platform generations is affordance.

The concept of affordance originates in ecological psychology. Gibson [3] proposed that organisms do not perceive their environment as a collection of objects with fixed properties. They perceive action-possibilities: what the environment affords the organism. A door handle affords gripping. A cliff edge affords a scenic viewpoint for a hiker, a jumping point for a daredevil, and a fatal fall for the unwary. The affordance is relational: it exists between what the environment makes available and what the organism can do with it. Norman [13] translated this to design: artifacts communicate possibilities for action independent of the designer's intent. Children can press a button before anyone explains what it does — the design affords the action. Hutchby [6] extended this to technology: artifacts have material properties that shape what users can do with them. However, affordances emerge from the interaction between what the artifact makes available and what the actor brings to it, not from the designer's intent alone. Bucher and Helmond [1] applied the framework to social media platforms specifically, distinguishing high-level affordances, or what the platform as a whole makes possible, from low-level affordances embedded in specific interface features.

The translation to this domain is direct. End-to-end encryption, designed to protect private communication for billions of legitimate users, affords unmonitored contact: communication that no platform can scan and no investigator can access without a court order. That affordance exists on WhatsApp, Signal, Telegram, and in the private message systems of every major platform. Timer-based content deletion, designed to make casual sharing feel lower-stakes, affords ephemerality: communication without a persistent record. On Snapchat, Instagram Stories, and WhatsApp, material sent disappears. In the enforcement record, this is evidence destruction by design. Account creation flows that require no verified identity, built to reduce onboarding friction, afford anonymity: action without accountability. File hosting and distribution infrastructure, built for sharing photos and documents, affords mass distribution of any content to anyone with a link.

Location-based discovery features, built to surface nearby users for social or dating purposes, afford proximity contact with strangers. Generative AI tools, built to expand computing possibilities and revolutionize content creation, afford the production of synthetic exploitative material, the fabrication of convincing personas at scale, and the automation of contact and grooming that previously required sustained human attention — capabilities that did not exist at this level of accessibility five years ago.

Different platforms. Expanding capabilities. Same function in the enforcement record.

Platforms are not designed to be exploitation-resistant. They are designed for engagement, retention, and scale [4]. The product decisions that produce each of these affordances are not made especially for offenders, or for any specific subset of the hundreds of millions of users. The capability exists for everyone. Misuse is an inherent consequence of building for everyone when the population includes people who have already decided to harm.

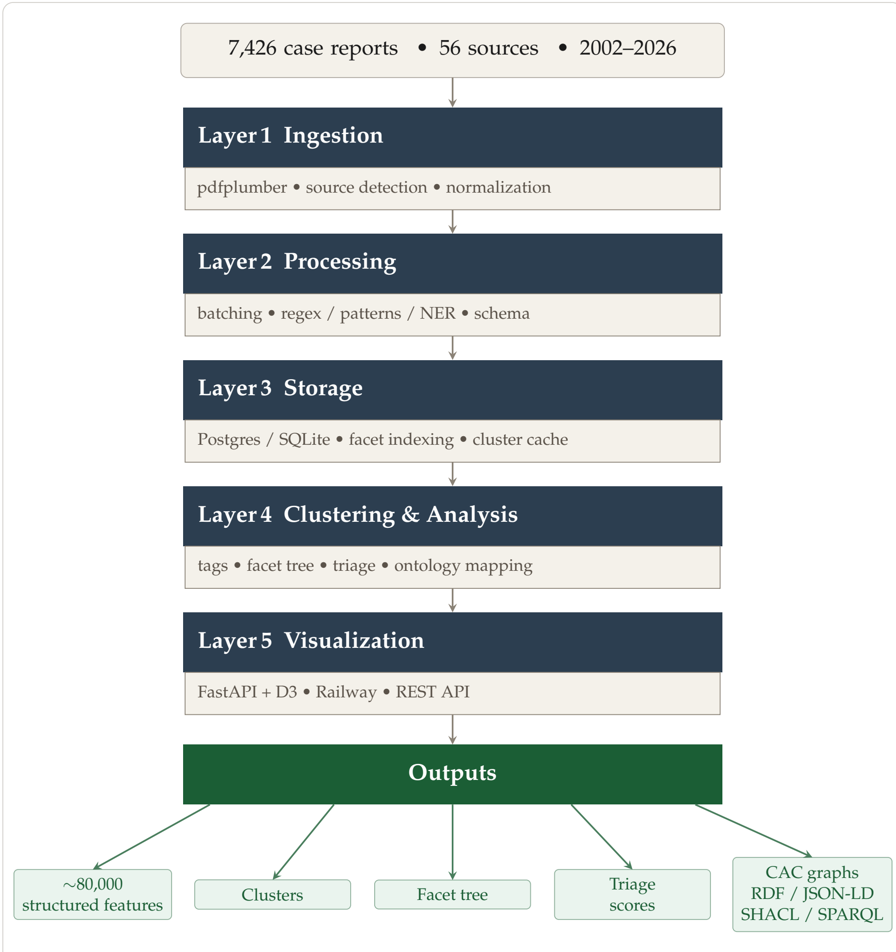
The affordance names what the platform offers, and the enforcement record provides evidence for how offenders misappropriated them. Section 3 presents both at scale.

3. Evidence at Scale: Analysis of 7,426 Exploitation Cases

3.1 Collection, Processing, and Validation

The corpus underlying this analysis consists of 7,426 publicly available Internet Crimes Against Children case records collected from 56 law enforcement sources spanning 2002 to 2026. Sources include task force press releases, agency annual reports, DOJ CEOS announcements, NCMEC public outputs, and federal court filings. All records are publicly available. No private case data, investigative files, or victim-identifying information is included in any form. The collection and analysis protocol operates in accordance with University of Massachusetts Amherst HRPO NHSR Determination #7668; this research does not contain private or personally identifiable information under federal regulations [45 CFR 46.102(f)(1),(2)].

Source selection targeted the full range of the U.S. ICAC enforcement ecosystem: 61 task forces, federal pipelines including Army CID, Customs and Border Protection, ICE/HSI, the U.S. Secret Service, and the U.S. Marshals Service, and produced 3,500 unique law enforcement agencies. Sources include Arizona ICAC Task Force, Texas Office of the Attorney General, Georgia Bureau of Investigation, and Michigan State Police, among dozens of others representing 24 years of enforcement activity.



CaseLinker processing pipeline — from raw enforcement records to queryable intelligence

Processing operates through a hybrid deterministic and interpretive machine-learning pipeline across three stages.

The first stage is regex-based extraction. Deterministic rules extract structured fields from case narratives across seven categories:

- **Perpetrator signals:** age, registered sex offender status, gender, and multi-defendant flags
- **Victim signals:** age, age ranges, stated victim count, and gender after validation
- **Relationships:** kin, role, or stranger when not stated
- **Prosecution:** charge phrases with counts and booking stage from arrest through sentencing; sentence durations

- **Evidence volume:** image, video, storage, and message counts when quantified in text
- **Platform identification:** named applications and surfaces — social media, messaging, gaming, file hosting, livestreaming, early-era chat clients, and generative AI tools when cited — alongside generic labels (online, chat, social media) when no product match is found
- **Technology signals** (stored separately from the platform list): investigation tooling (PhotoDNA, hash-matching, CyberTipline language), anonymization infrastructure (Tor, dark web, cryptocurrency), and P2P clients

The second stage is pattern-based classification. Each case receives topic labels (production, possession, distribution, trafficking, CSAM, AI-generated CSAM, sextortion, hands-on versus online-only, family versus stranger, international, multi-state) and severity indicators (infant, very young, under 12, sexual abuse, multiple perpetrators). Severity phrases used in triage priority scoring are extracted separately: language indicating active, ongoing, or escalating abuse is weighted in the priority model.

The third stage is ML enrichment. When the ML stack is enabled, NER via Stanford NLP (Stanza) adds organizations, locations, dates, and ages, merged with regex output under a precedence model that preserves deterministic extractions when conflicts arise. A victim-age gate drops decoy and headline ages introduced by undercover investigation language. Semantic sentence scoring attaches concept weights to each case; grooming severity tags and strong possession or AI-generation language reinforce corresponding case topics. Concept scores are retained for downstream analysis; a curated subset are merged into main fields.

Extraction is context-aware throughout, distinguishing investigation types, victim from perpetrator gender, and perpetrator admissions where explicitly present. Agency normalization resolves over 3,796 unique agency string variants to canonical identifiers using a combination of pattern matching and NER, enabling cross-source aggregation across jurisdictions that name the same agency differently across releases.

The primary evidence base for platform harm analysis (Q1) is a candidate pool selected by automated filter, not manual case selection. A case enters the pool if its structured metadata contains at least one non-generic platform label, excluding generic placeholders ("online," "internet," or generic tags). This yields 1,875 candidate cases (25.1% of the corpus) spanning 54 named platforms; the remaining 74.9% lack a named platform and fall outside affordance-level analysis due to insufficient signals for supporting claims at the level of a specific platform capability. This is partly a property of the source material itself: public case documents systematically under-report platform and offense detail to avoid influencing offending behavior, compromising ongoing investigations, or revealing prosecutorial strategy (discussed in Section 7.5). Restricting Q1 to cases with a named platform trades corpus breadth for evidentiary precision as affordance-level analysis requires knowing which platform, and which of its features, was actually present.

Each candidate is processed by an evidence extractor that strips PSA and awareness-campaign boilerplate, extracts a supporting quote, assigns an offense role, and classifies evidence strength at the platform–case level. The candidate-to-evidence pipeline yields a final Q1 evidence base of 1,875 cases across 3,128 platform–case records.

Evidence strength is tiered. *Stated* records link a platform instrumentally to offense conduct in the same sentence — through an explicit construction (*through*, *via*, *using* a platform) or an offense keyword in close proximity, excluding awareness framing and enumerated app lists. *Inferred* records show platform and offense language co-occurring without instrumental linkage. *Named-only* records carry a metadata platform tag without qualifying textual support. At the case level, 858 cases (45.8%) contain at least one stated record, 136 (7.3%) carry inferred-only evidence, and 881 (47.0%) are named-only. The affordance labels applied in Q1 are a manual analytical layer over this automated extraction: the pipeline supplies quotes, roles, and tiers; affordance semantics are assigned by hand.

The full pipeline — extraction rules, classifier weights, normalization dictionaries, and reproducibility documentation — is released as open-source software under the MIT License at github.com/mrinaalr/CaseLinker.

3.2 Ontologies for Modeling Radioactive Data

CaseLinker extracts over 80,000 structured features across 10 dimensions and stores thousands of cases in a PostgreSQL database for analysis. However, tabular representations are insufficient for the questions this paper asks. A case record is not a row. It is a set of relationships: an offender reached a victim through a platform, used a capability to produce material, and distributed it through infrastructure. A relational table can store the tags, actions, and data. It cannot natively represent the structure and relationships between fields. The questions Q1's affordance-based platform harm analysis asks — which capabilities cluster with which offense types, which platform features co-occur across cases, how offense mechanics distribute across a 24-year enforcement record — are relational questions. They require a data model that treats relationships as first-class objects.

Knowledge graphs provide that model. Each case in this corpus is represented as a directed, typed graph: nodes for entities (offenders, victims, platforms, charges, agencies), edges for relationships (`contacted_via`, `charged_under`, `investigated_by`), and typed properties for attributes. This representation enables the pattern queries Q1 requires: offense events, platforms, agencies, and actors as pieces of an interconnected system, not isolated records.

The vocabulary that models these cases is the Crimes Against Children (CAC) Ontology, shepherded by Project VIC International as an interoperable standard for representing crimes-against-children investigations. The stack is layered: gUFO provides foundational ontology primitives. UCO (Unified Cyber Ontology) models cyber investigation objects and relationships. CASE (Cyber-investigation Analysis Standard Expression) defines how investigation narratives are expressed as knowledge graphs. CAC extends that stack for crimes against children specifically: platforms, victims, offenders, investigations, and outcomes as typed entities rather than free text.

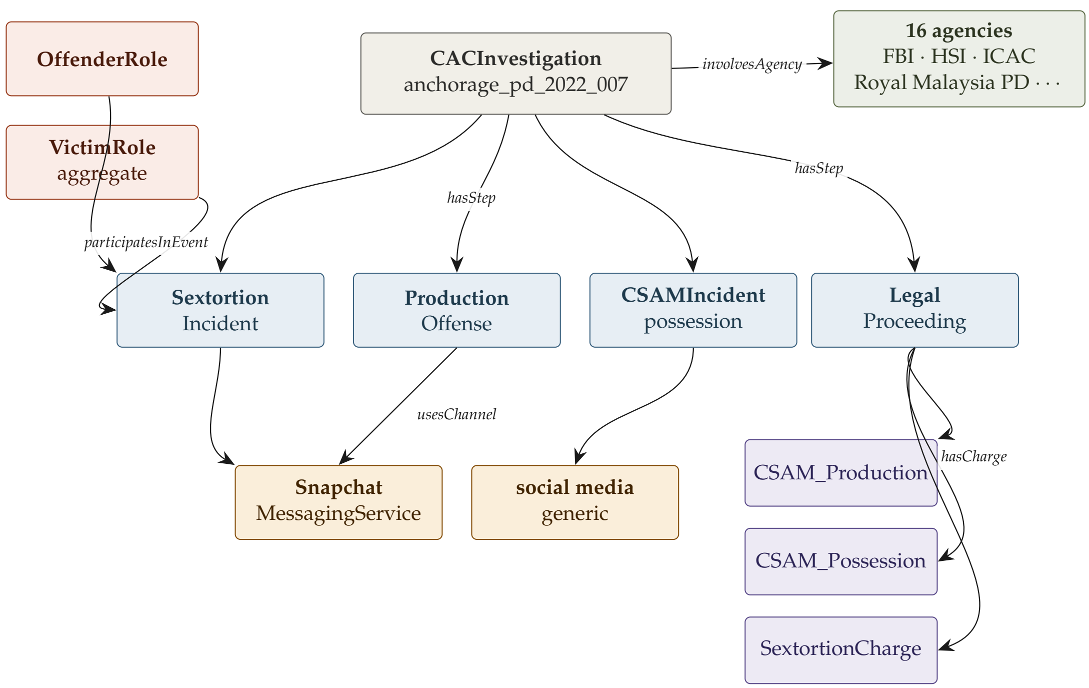


Figure 1: CAC ontology representation of a single case as emitted by the `case2cac` pipeline (case `anchorage_pd_2022_007`).

CaseLinker maps each extracted case feature to a CAC class. Platform mentions become typed `cac:Platform` nodes with semantic relationships to offense events. Statutory charges map to identifiers linked to offense type taxonomies. Perpetrator behaviors map to grooming stage classes. The mapping is explicit and auditable:

every ontological assertion traces to a source extraction and a source text span. There are no black-box inferences. Structural validation is performed using SHACL (Shapes Constraint Language) against a curated subset of the CAC module stack; each case graph is validated prior to inclusion in the Q1 evidence base, and non-conforming graphs are flagged and excluded rather than silently retained.

The result is a corpus of 1,800+ SHACL-validated case graphs forming the evidence base for Q1 analysis. Graph construction at corpus scale is handled by the CaseLinker MCP (Model Context Protocol) server: 37 tools spanning case2cac graph construction, cross-case concept search and comparison, traversal, validation, and export across press-release and case-record sources.

Primary-source court record representations — dockets, statements of offense, and federal indictments retrieved from primary court filings — are handled by the CASE-UCO SDK directly. The CASE-UCO SDK is a typed, multi-language library (Python, C#, Java, Rust) implementing 428+ ontology classes across the CASE/UCO standard, designed for constructing and validating JSON-LD knowledge graphs for digital forensics and cyber-investigation workflows. Where the case2cac tools operate over structured fields at scale, the CASE-UCO SDK is used for precision representation: extracted facts from primary federal court documents are mapped to typed ontology classes, producing ground-truth graphs that anchor corpus-level pattern analysis.

3.3 Research Questions and Analytical Approach

Three empirical and one theoretical question organize the analysis. Each draws on a distinct evidence layer: the 7,426-case corpus for breadth, the 1,800+ validated knowledge graphs for structured pattern analysis, and 30 federal court records (indictments, plea agreements, and defendant-stipulated Statements of Offense) retrieved from PACER for ground-truth anchoring.

Q1 — Platform Affordances. Which platform capabilities do offenders exploit most consistently, across case records, offense types, and the full period this corpus covers? Q1 operates at the platform level, drawing on every case with at least one identified platform in the enforcement record. The 30+ platform affordance table maps each platform to its documented misuse surfaces and harm vectors across stated, inferred, and named-only evidence tiers. The knowledge graph representation enables this analysis without reducing to frequency counts: a query across the graph can ask which capability types co-occur with which offense types, which platforms share affordance profiles, and how those profiles shift across offender typologies and agencies the corpus represents.

Q2 — Exploitation Life Cycle. How is technology implicated across the four offense subtypes — grooming and enticement, sextortion, CSAM production, and coordinated criminal networks — from first contact through prosecution? Q2 operates at the harm-signature level. The primary evidence base is four federal district court cases, retrieved from Public Access to Court Electronic Records (PACER) and modeled via the CAC Ontologies:

- *United States v. Rehman* (D.D.C. 1:23-cr-00064) — §2422(b) enticement; platform-mediated grooming and relationship formation
- *United States v. Amin* (D. Alaska 3:22-cr-00055) — 13-count §2252A(g) enterprise; sextortion at scale across more than 80 Snapchat and 40 Instagram accounts
- *United States v. Pathmanathan* (D.D.C. 1:22-cr-00150-JEB) — §2251(a),(e) production; Instagram-to-Messenger platform migration and live stream directed self-production
- *United States v. Bermudez et al.* (E.D.N.Y. 1:25-cr-00361) — six-defendant §2252A(g) coordinated network; Discord and gaming platform infrastructure

An additional twenty-six federal prosecution records — indictments, plea agreements, Statements of Offense, trial briefs, government sentencing memoranda, and findings of fact — were captured through PACER requests after a blend of targeted and random traversal of CaseLinker's facet tree. Each case was mapped to the CAC ontology independently of the automated pipeline, anchoring corpus-level patterns against primary federal court records.

Q3 — Disruption. Where in the platform stack and enforcement pipeline do realistic opportunities to disrupt exploitation exist? Q3 synthesizes Q1 and Q2: given which capabilities offenders consistently reach for, and how those capabilities combine into recognizable offense mechanics, which affordances are most amenable to design-level, policy-level, or enforcement-level intervention? Q3 is the operational output of the analysis, the translation from documented pattern to actionable intervention.

Q4 — Formal Framework. Can the relationship between platform affordances, offender goals, and victim-facing harm be expressed in a generalized mathematical model for exploitation? Q4 operates at the level of abstraction, formalizing the patterns the corpus documents. The mathematical framework in Section 7 maps offender goals to affordance trajectories (φ), exploitation types to harm sets (η), and affordance classes to co-occurring harms (ψ) — producing a marginal exploitation utility measure $u(a_{\text{new}}, g)$ that is estimable before a feature enters the enforcement record.

Q1 characterizes the affordance landscape. Q2 anchors it in the court-verified mechanics of four distinct offense types. Q3 asks what can be done. Q4 builds the vocabulary to reason about exploitation before it occurs.

Practitioners and policy readers may proceed directly to Section 4 for the platform analysis and Section 6 for intervention recommendations; the formal framework in Section 7 is self-contained and can be read independently.

4. Platform Affordances and Misuse Surfaces

The enforcement record names 54 distinct platform labels across the Q1 evidence base. Analyzed individually, they appear chaotic: products from different eras, built by different companies, serving different purposes. Analyzed by affordance, they resolve into a small number of recurring signatures.

Table 1 presents the platform manifest: every analyzed platform with sufficient case representation, its affordance signature, the misuse surface that offenders utilize, and the victim-facing harm vectors documented in the corpus. Three definitions discipline the table. An **affordance** is a neutral capability the platform's design makes possible. A **misuse surface** is how an offender turns that affordance toward an offense; it is offender-facing. A **harm vector** describes what happens to a child; it is victim-facing, drawn from a fixed set established across the corpus, and ranked by documented prevalence. [1]

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Messaging Services		
Kik 208 stated · 352 total	*Anonymity, group discovery.* Pseudonymous contact without traceable identity; closed-group trading; account churn to evade detection.	Child's abuse imagery distributed/traded (re-victimization) Child contacted through anonymous messaging Child groomed toward contact

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[1]The manifest measures *documented offender use of platform capabilities*, not ground-truth harm frequency. Possession and distribution of CSAM dominate the record partly because they are the most detectable evidence of harm and play a significant charging role after being surfaced through the CyberTip pipeline; contact and grooming are systematically under-detected and under-documented in public records. Affordances and misuse surfaces are derived from analysis of each platform's capabilities; harm vectors are drawn only from offenses the corpus actually documents. Platform counts are a lower bound on involvement, not a complete picture.

Table 1 (continued)

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Snapchat 169 stated · 257 total	*Ephemerality, camera-first capture.* Evidence destruction by design; real-time coercion of imagery; ephemeral solicitation with reduced forensic trail.	Child contacted and exploited through ephemeral messaging Child's abuse imagery distributed/re-shared (re-victimization) Child groomed toward contact
Discord 43 stated · 99 total	*Server/community structure, persistent contact.* Closed servers as low-visibility spaces; coordination among offenders; sustained contact enabling coercion.	Child's abuse imagery distributed/traded (re-victimization) Child contacted through servers and DMs Child sextorted under threat Child groomed toward contact
Social Media Platforms		
Facebook 55 stated · 274 total	*Digital profiles; friend, network, and mutual-connection discovery; search by name, school, location, interests.* Profile-mining to target; fake/impersonation accounts to approach minors under false identity; migration target — contact initiated on Facebook, moved to private/ephemeral platform.	Child's abuse imagery solicited / produced / distributed Child contacted through profiles and Messenger Child groomed toward contact
Instagram 27 stated · 174 total	*Visual-first profiles, DM, discoverability.* Visual profiles to select targets; DM coercion; disappearing messages reduce forensic trail.	Child contacted through DMs and visual profiles Child's abuse imagery distributed/re-shared Child sextorted under threat
Reddit 10 stated · 22 total	*Image and video upload to platform accounts, account-linked identity, public posting surface.* Personal account CSAM upload and post for distribution and re-sharing; forum posting surface used to initiate contact with minors.	Child's abuse imagery distributed, traded, or re-shared (re-victimization)
TikTok 4 stated · 25 total	*Algorithmic discovery, DM, public-by-default.* For You feed surfaces minors without active search; public videos to select targets; DM to move contact off the public feed.	Child contacted and exploited through messaging Child groomed toward contact or meeting Child solicited for explicit imagery
File Hosting Services		

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Table 1 (continued)

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
Dropbox 33 stated · 86 total	*Cloud storage, public link sharing, desktop sync.* Shared links distribute CSAM collections without direct transfer; desktop sync client mirrors local library to cloud automatically; handoff layer — link shared via separate contact channel.	Child's abuse imagery possessed or stored in offender's cloud account Child's abuse imagery uploaded for distribution or collection Child's abuse imagery distributed or re-shared via shared links (re-victimization)
Mega.nz 3 stated · 15 total	*Cloud storage, shareable links to hosted collections, cross-platform application access.* Distributed CSAM collections; handoff layer in multi-platform distribution stacks — link shared on Snapchat, Kik, or fetish sites, recipient downloads the Mega app to retrieve material;	Child's abuse imagery distributed or re-shared via cloud storage links (re-victimization) Child's abuse imagery possessed or stored in offender's cloud account Child's abuse imagery uploaded to cloud file hosting for distribution or collection
Anonymous Chat Platforms		
Whisper 3 stated · 11 total	*Anonymity, location-based discovery.* Anonymous contact with minors without traceable identity; location proximity used to surface nearby targets; DM moves contact off public feed into private channel.	Child contacted and exploited through anonymous messaging Child groomed or enticed toward sexual contact or meeting Child solicited for explicit imagery through anonymous posts or direct messages
Omegle 2 stated · 7 total	*Anonymous stranger pairing, unmonitored video.* Random pairing reaches minors without targeting effort; zero-identity architecture leaves no forensic trail; unmonitored video channel used for real-time solicitation and exhibition; migration target to persistent platforms.	Child contacted and exploited through anonymous random video or text chat Child groomed or enticed toward sexual contact or meeting Child solicited for explicit imagery or subjected to real-time sexual exhibition in video chat
AI Services		
Gen AI 21 stated · 54 total	*Generative synthesis, likeness manipulation.* Synthetic CSAM from prompts; nudification of real photos; model fine-tuning on a target's images.	Real child's likeness manipulated/synthesized into abuse imagery without contact Child's abuse imagery (incl. AI-generated) distributed/possessed
Additional Surfaces		

continued on next page

Table 1 (continued)

Platform	Affordance Signature / Misuse Surface	Documented Harm Vector
P2P (BitTorrent) 3 stated · 8 total	*Peer-to-peer file sharing, mini-server distribution from central device, network search by keyword.* CSAM collections downloaded and saved from the network; files exchanged or traded with other users; offender's computer made available as a distribution node, seeding material to any peer who requests it;	Child's abuse imagery distributed, traded, or re-shared (re-victimization)
Video Streaming 6 stated · 28 total	*Real-time bidirectional video in private chat or 1:1 session, live remote viewing of another party's camera feed, streaming media capture.* Coercion into production and performance of explicit acts during a live private session; session recorded — including without victim knowledge — to manufacture CSAM; captured recording used as coercion leverage to demand more explicit conduct under threat of exposure.	Child solicited for explicit imagery or subjected to real-time sexual exhibition in video chat Child sextorted under threat after session imagery obtained Child contacted and exploited through platform messaging
Gaming Platforms 4 stated · 29 total	*Interactive user community, built-in chat, low-friction contact.* In-game chat used to initiate contact with minors under cover of shared gameplay; pseudonymous accounts with no identity verification; migration target — contact initiated in-game, exploitation moves to Kik, Whisper, or anonymous messaging platforms.	Child contacted and exploited through in-game chat or messaging Child groomed or enticed toward sexual contact or meeting Child's exploitation migrates off-platform to anonymous messaging after initial in-game contact

Full analysis and supporting evidence for all 54 documented platforms available on the platform harm dashboard.

4.1 Contact and Approach Affordances

Table 1 organizes the enforcement record by platform. This section and the three that follow reorganize the same record by affordance class, at the level of which patterns hold across platform generations.

Contact and approach is the foundational class. Every victim-facing offense type this corpus documents begins here. An offender cannot groom a child they have not reached. The imagery acquisition that makes sextortion possible requires a channel to the victim before the first image changes hands. Directed production requires sustained contact over a medium capable of video. Coordinated enterprises recruit through contact before they produce, distribute, or coerce.

Across 1,720 cases in the Q1 evidence base, contact and approach affordances materialize in six recognizable forms. Two reduce targeting effort. Four extend reach without mandated accountability.

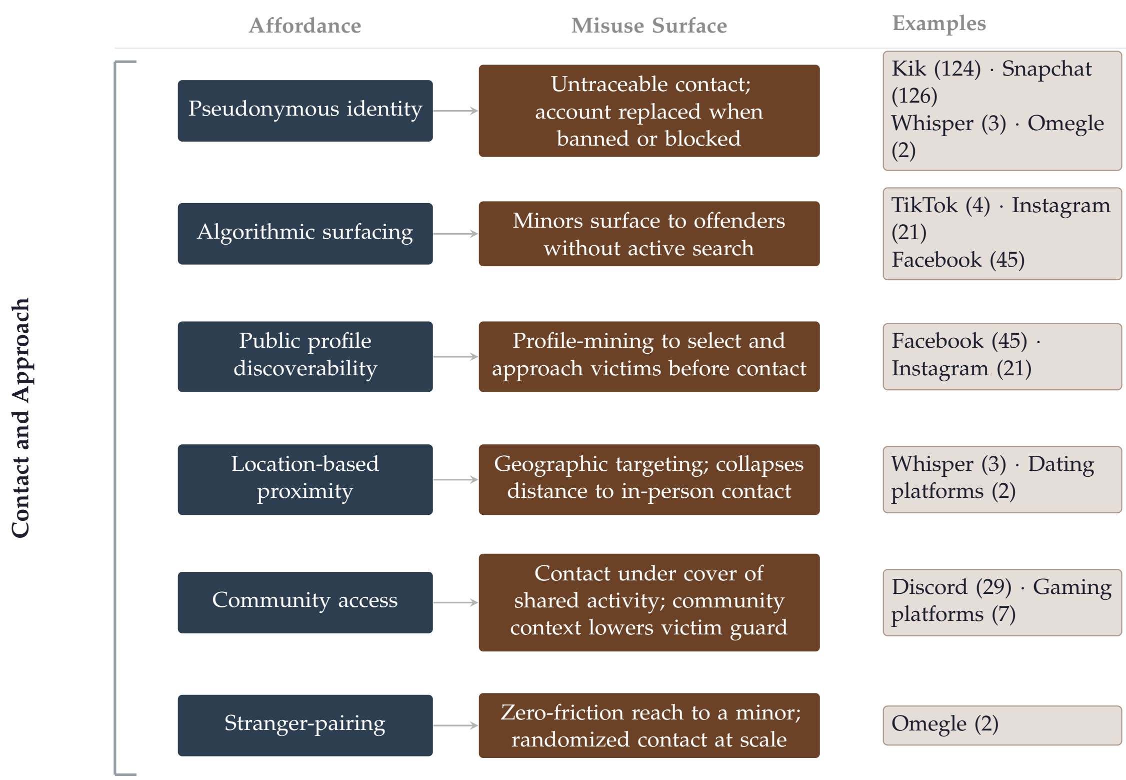


Figure 2: Contact and approach affordance implementations. Flow traces from implementation type through platform mechanism to the misuse surface created in the enforcement record.

4.2 Production Affordances: The Generative Evolution

Production is the class of affordances that enable abuse material to be created. The material does not preexist the offense. The offender engineers its existence: directed remote capture, coerced performance on camera, or generative synthesis from a text prompt or manipulated photograph.

Generative AI introduced the first production pathway in this corpus that does not require the contact class. Every preceding production offense required a victim present during creation: a live video channel to direct, a messaging interface to receive self-generated imagery, or physical proximity to film. The child's participation was the necessary condition. Generative AI removed that condition for the first time in the enforcement record. A text prompt now produces material without presence or compliance. A real child's photograph, run through a locally-deployed model, becomes abuse imagery without any contact event. The harm of production — a child's likeness, captured and circulated — no longer requires the child to be present when the material is made.

Across 177 cases in the Q1 evidence base, production affordances materialize in five recognizable forms. The first three require a victim. The last two do not.

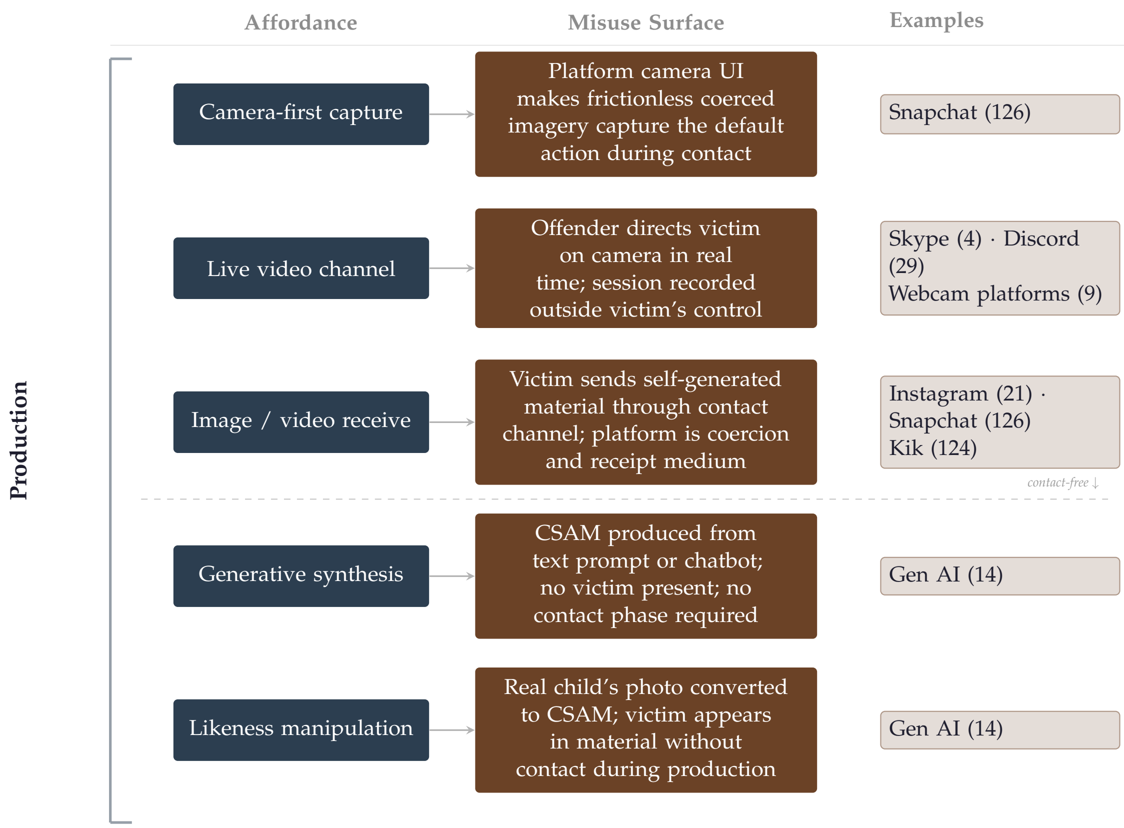


Figure 3: Production affordance implementations. The dashed line marks the generative threshold: affordances above it require a victim present during production; affordances below do not.

4.3 Possession and Trade Affordances

Possession and trade is the class of affordances that enable the storage and exchange of abuse material. Contact reaches the victim. Possession and trade extend the harm past the offense event. Material captured in production does not disappear when the session ends. It accumulates, organizes, and moves. A shared link distributes a collection to parties the offender never contacted. A closed trading group multiplies re-victimization across a network without requiring additional access to the original victim. The offense does not end when contact does, harm perpetuates wherever the material lands.

Across 1,708 cases in the Q1 evidence base, possession and trade affordances materialize in five recognizable forms. Four address movement. Two address storage.

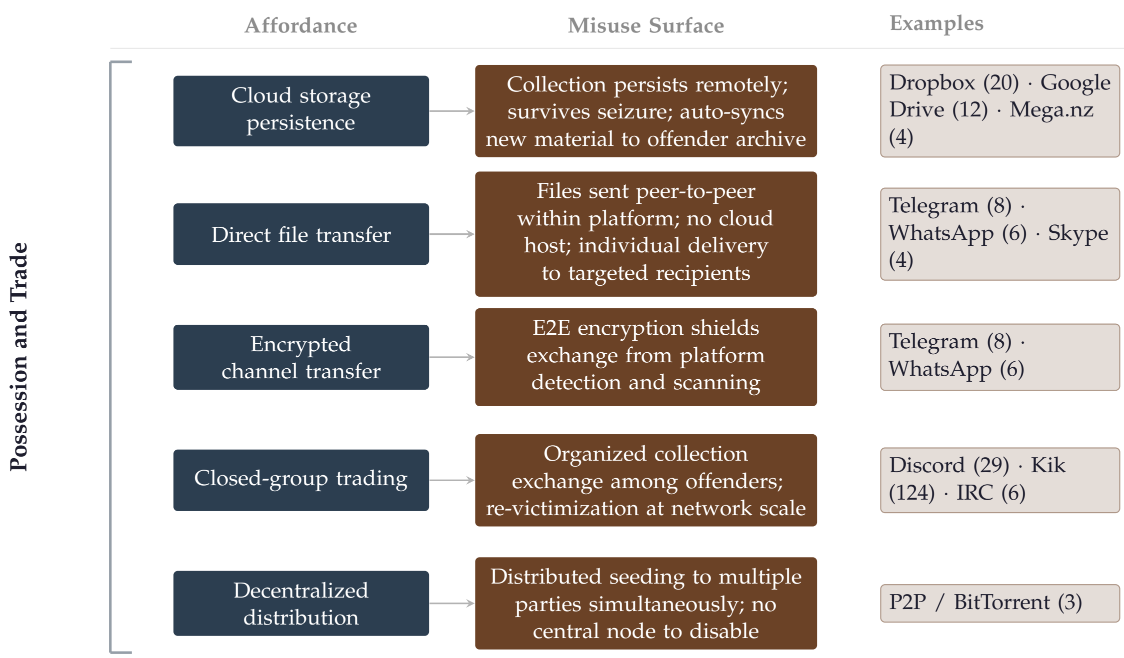


Figure 4: Possession and trade affordance implementations. Flow traces from affordance through the misuse surface it creates in the enforcement record.

4.4 Coordination Affordances

Coordination is the class of affordances that enable exploitation at enterprise scale. Contact, production, and possession can each be executed by a single offender pursuing their own exploitative goal. Coordination is what allows multiple offenders to pursue those goals simultaneously, across specialized roles, against multiple victims at once. Coordination affordances keep the individual offense mechanics similar and multiply the hands executing them.

Across 198 cases in the Q1 evidence base, coordination affordances materialize in five recognizable forms. Three concentrate in Discord's server infrastructure. Two extend to gaming platforms.

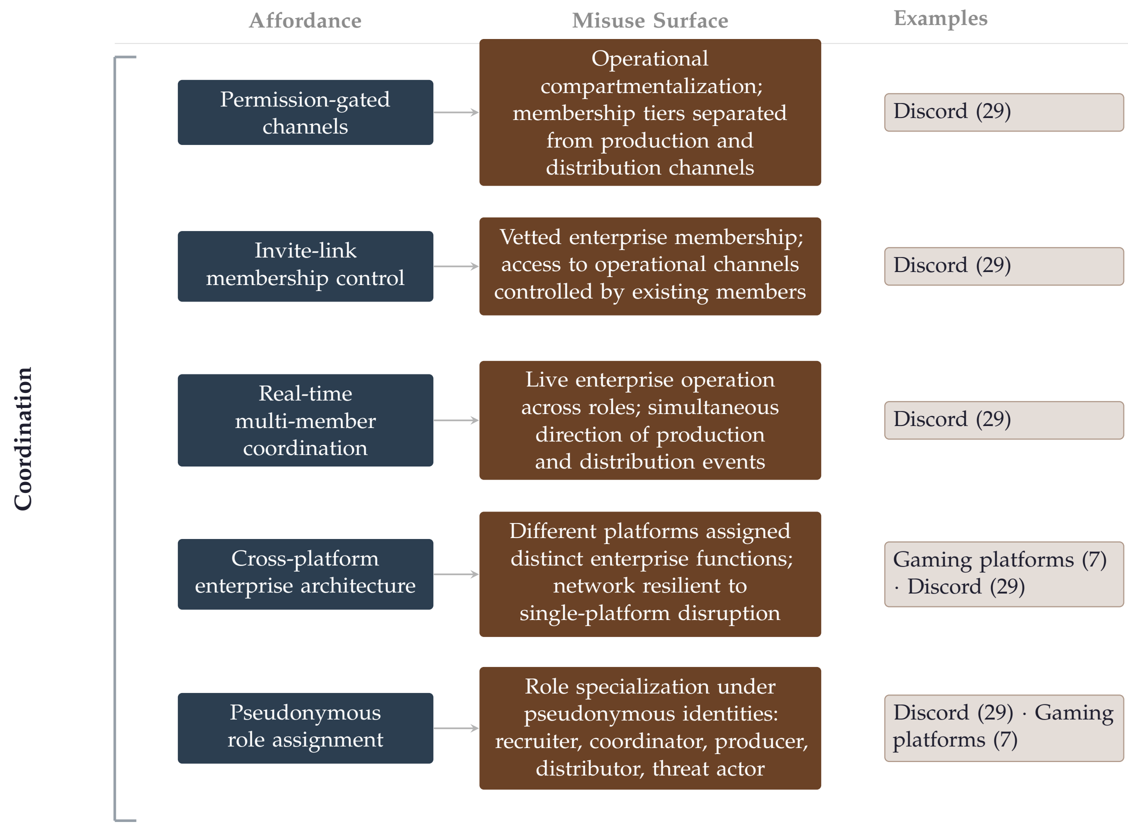


Figure 5: Coordination affordance implementations. Flow traces from affordance through the misuse surface it creates in the enforcement record. Case counts at right are lower bounds on platform involvement; see Section 3.1.

The four affordance classes this section documents establish what the platforms in this enforcement record make possible and the misuse surfaces they expose. Section 5 explores how offenders weaponize those affordances.

5. Harm Signatures: How Technology Shapes Exploitation

The platform manifest in Section 4 maps capabilities to misuse surfaces. It answers a static question: what does each platform make possible? This section asks the harder one. Across four offense subtypes — grooming and enticement, sextortion, the production of abuse material, and coordinated criminal enterprises — how do those affordances actually combine during the mechanics of an offense, from first contact through victim-facing harm? Each subsection develops a **harm signature** for its offense type: a generalized exploitation lifecycle mapping the steps an offender takes at each stage to the victim-facing harms those actions leave behind. The evidence base is four federal district court cases retrieved from PACER and read in full, each mapped to the CAC ontology independently of the automated pipeline. They are ground-truth anchors within the same analytical framework: primary-source court records that document, in precise legal language, exactly which capabilities were used, in which sequence, to reach which victims.

5.1 Grooming and Enticement

Grooming is applied contact. The affordances Section 4.1 documents initiate every harm signature and grooming dynamics run through every victim-contact offense type in this section. The sustained contact that makes production possible is grooming. The deception and contact that make sextortion's first image possible draw on the same mechanics as grooming. The recruitment mechanics that allow a coordinated enterprise to scale across victims are grooming. What changes across offense types is what the offender does after the relationship is established. The relationship itself is always the same structure: a child brought, through sustained deception or threats, to a state of compliance they would not have reached otherwise.

The enticement-to-contact trajectory is also the most probabilistic of the four. A sextortion offender who holds retained imagery has a decisive advantage that does not depend on the victim's continued willingness. An enterprise who has already directed explicit conduct has created evidence that binds the victim through fear. A grooming offender has none of that leverage in the early stages. Every request is a gate. Every gate is a compliance decision the victim can refuse. What the offender is managing, across every contact and every escalation, is a probability distribution over victim responses — and the grooming mechanics exist precisely to shift that distribution. Flattery raises the probability of the next compliance. Reciprocal disclosure lowers the victim's sense of asymmetry. Persistence reduces the exit window. Exploitation of disclosed vulnerability — depression, isolation, prior trauma — narrows the space between compliance and refusal.[2]

The chain documented in this case runs from first contact through flattery and low-stakes requests, through escalating solicitation and directed production, to meeting arrangement and hands-on contact, all within approximately five weeks on a single platform. At no point in this chain did the offender use threats, quotas, or distribution leverage. Compliance was obtained entirely through relationship

[2] *United States v. Rehman*, D.D.C. 1:23-cr-00064-CJN, Statement of Offense (filed Nov. 21, 2024). Structural observations in this subsection derive from the Statement of Offense read in full. No victim-identifying information appears here.

investment: sustained presence, flattery calibrated to disclosed vulnerability, and the normalization of escalating requests through reciprocal disclosure. Each compliance gate passed lowered the probability of exit at the next one. The trajectory did not require coercion in the technical sense because the relationship cultivation had already done the coercive work before any explicit demand was made.

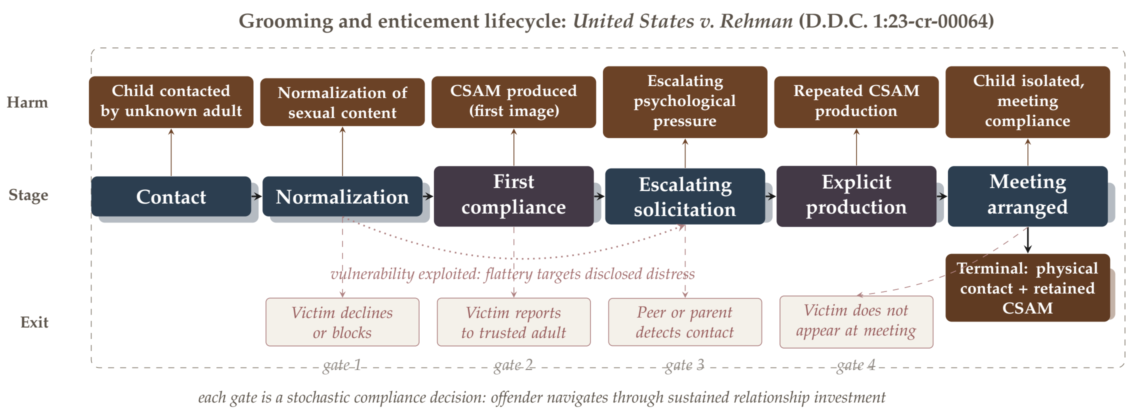


Figure 6: Grooming and enticement lifecycle: *United States v. Rehman* (D.D.C. 1:23-cr-00064) mapped onto the generalized exploitation lifecycle. Gate nodes (purple) mark stochastic compliance thresholds where the trajectory could have collapsed. Exit branches below the spine show documented disruption opportunities at each gate. The vulnerability arc denotes offender adaptation to disclosed victim distress across the escalation phase.

The terminal harm state here is physical contact, not perpetual online coercion. This distinguishes the enticement-to-contact trajectory from sextortion, where the terminal state is retained material held under constant threat of release. In this case, the CSAM produced during the online phase is a precursor to the contact goal — a stage in escalation, not the instrument of continued exploitation. Four documented exit points existed in the chain. The offender navigated all four through the same mechanism: sustained relationship investment targeted at a child whose vulnerability had been disclosed and whose trust had been deliberately cultivated.

5.2 The Production of Abuse Material

Production is the offense of directed creation of CSAM. The material does not preexist it. The offender engineers its existence by coercing a victim to perform explicit conduct on camera, or by filming in-person abuse directly. The act of exploitation becomes a permanent artifact the moment capture occurs. The lifecycle proceeds in two phases.

Remote, online-directed capture is the dominant production modality in this corpus. The first phase is access. The offender establishes contact on a public surface and escalates through the grooming and social engineering pathways documented in Section 5.1. Communication then shifts to a channel capable of image or video production. The contact surface is for access. The production channel is for capture.

The second phase is directed production. The offender issues commands during the live session. The victim performs them on camera. The offender records the session.[3] When the victim resists, the offender threatens to distribute previously captured material or escalates threats and compliance resumes. When a victim blocks contact entirely, the offender creates a new account and returns to the contact stage. The lifecycle does not break. It resets.

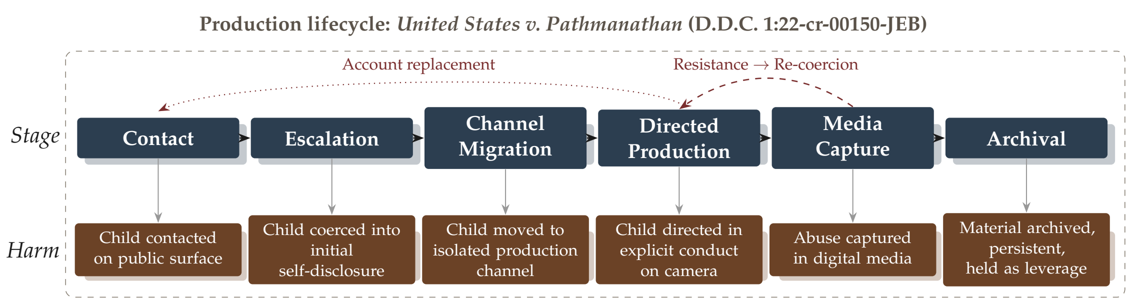
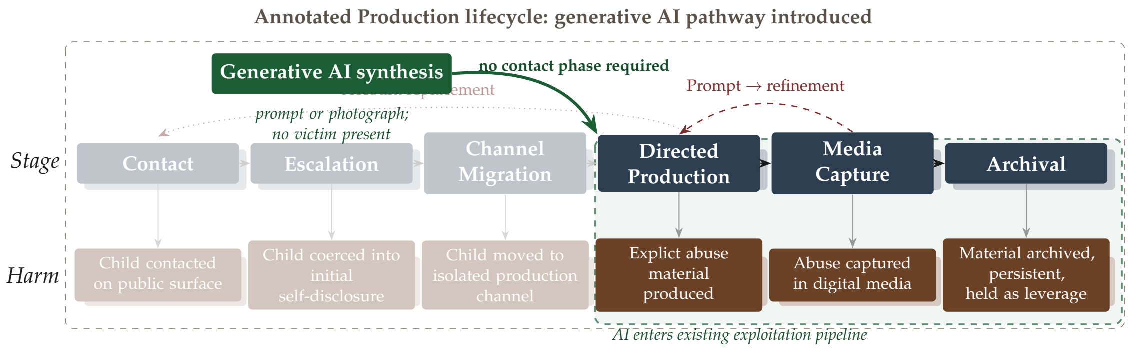


Figure 7: Production lifecycle: *United States v. Pathmanathan* (D.D.C. 1:22-cr-00150-JEB) mapped onto the generalized exploitation lifecycle. Dashed arc denotes the resistance and re-coercion loop within an active session; dotted arc denotes account replacement when a victim blocks contact entirely.

Generative AI has introduced a distinct pathway that Figure 7 does not map. Section 4.2 documents how generative synthesis removes the contact phase as a precondition. Synthesized material, regardless of its statutory classification, yields media that can be used to reinforce established exploitative harm signatures through normalization in grooming sequences, coercive leverage in sextortion chains, and compliance priming prior to directed production.

[3] *United States v. Pathmanathan*, D.D.C. 1:22-cr-00150-JEB, Statement of Offense (filed Jan. 30, 2026). Structural observations in this subsection derive from the Statement of Offense read in full. No victim-identifying information appears here.



The production harm signature terminates in an archive. Material captured across sessions is organized and held outside any channel the victim can access. Each session adds to it. The archive outlasts every session and serves as a permanent record of the crimes committed.

5.3 Sextortion: Coercion After Contact

Sextortion is distinct from every other offense type in this corpus. The harm does not begin with the production of material, it begins before, in the act of deception and coercion that makes production possible. And unlike grooming, which moves toward a physical meeting, or production, which terminates when material exists, sextortion is self-sustaining: the material produced in the first exchange becomes the instrument of the next. The chain feeds itself. Every image a victim sends under threat becomes leverage for demanding another.

The mechanism is coercion under retained threat. A first image is obtained — through impersonation, false promises, or social engineering. Once that image exists, the offender's position is asymmetric: the victim cannot undo the transfer, and the offender controls whether it spreads. That asymmetry is the engine of the offense. Threats to expose the image to the victim's social network — friends, family, schoolmates — convert a single act of exploitation into a sustained compliance mechanism. The victim is not coerced once. They are coerced continuously, each demand backed by the same retained material and the constant threat of its release.[4]

What sustains the loop is the same threats, redeployed. Daily quotas of images and videos, directed conduct on live calls, demands for increasingly explicit material: each round of compliance produces new material, which extends the leverage,

[4] *United States v. Amin*, D. Alaska 3:22-cr-00055-SLG-KFR (indictment filed Oct. 5, 2022). Structural observations in this subsection derive from the federal indictment read in full. No victim-identifying information appears here.

and enables the next round. The offender and victim's positions grow increasingly asymmetric, as retained material reinforces the coercion loop. Platform bans do not break the chain. When accounts are disabled, new ones are created and victims are told to reconnect. The coercion loop persists across platform interventions because the leverage — the retained imagery — exists outside any single account.

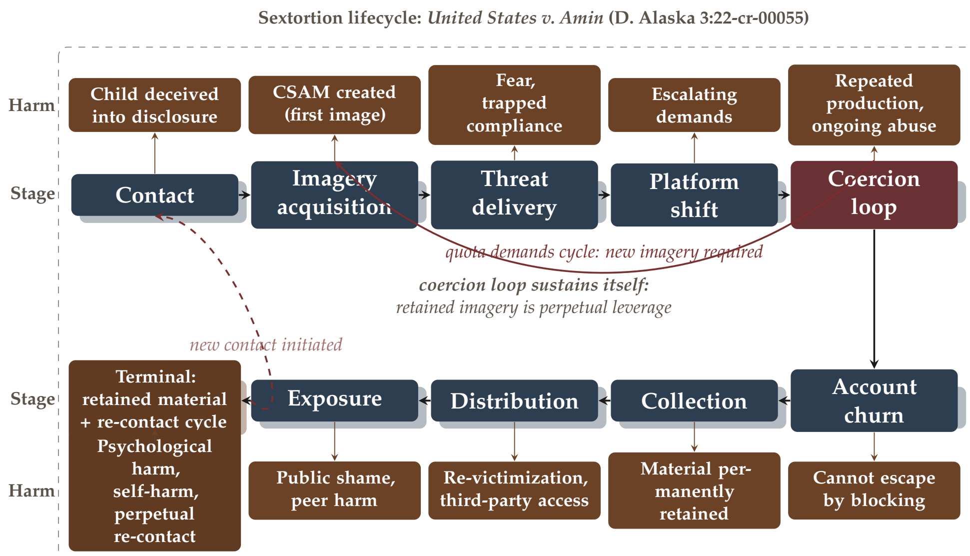


Figure 8: Sextortion lifecycle: *United States v. Amin* (D. Alaska 3:22-cr-00055) mapped onto the generalized exploitation lifecycle. The coercion loop (from threat delivery back to imagery acquisition) is the defining mechanic: retained material is perpetual leverage.

The lifecycle documented in this case extends beyond the coercion loop into collection, distribution, and exposure. Material is organized and stored; links are shared with co-conspirators who must harass victims before receiving access. Victims' images are sent to their friends and schoolmates when they attempt to stop. And the chain propagates: victims under active coercion are directed to recruit other minors, supply follower lists, and warn peers that they too will be exposed. The sextortion lifecycle does not terminate when the offender stops. It terminates, if it terminates at all, when the retained material is no longer reachable or the offender is apprehended.

5.4 Coordinated Criminal Enterprises

The preceding three subsections each describe a single offender pursuing a goal through a sequence of actions. Grooming and enticement require sustained contact and trust architecture. Sextortion requires imagery acquisition and coercion leverage. Production requires a victim, a camera, and a channel. Each is the work of a single offender. Coordinated criminal enterprises are distinct in one respect: the trajectory is parallelized across actors. The affordance set is the same. What changes is that no single offender executes the full sequence. Recruitment, production, distribution, and coercion become roles, each filled by a different member of the enterprise, each operating on a different phase of the same trajectory simultaneously.

The coordination affordance makes this possible. Discord's server and channel architecture, designed for community organization and role-based access control, functions as the enterprise's headquarters. [5] Locked channels become internal operational security: the same permission system that lets a gaming community separate beginner and advanced discussion lets an enterprise separate its membership from its operations. Multiple-platform architecture — gaming platforms as recruitment funnel, communication channels as operational headquarters — is a key signature of exploitation that has scaled past what any single offender can execute.

What the enterprise structure enables that solo offending cannot is scale and role specialization. A solo offender who produces, distributes, and coerces is executing each phase sequentially, bounded by individual bandwidth. An enterprise distributes those phases across members, executing them in parallel across multiple paths simultaneously.

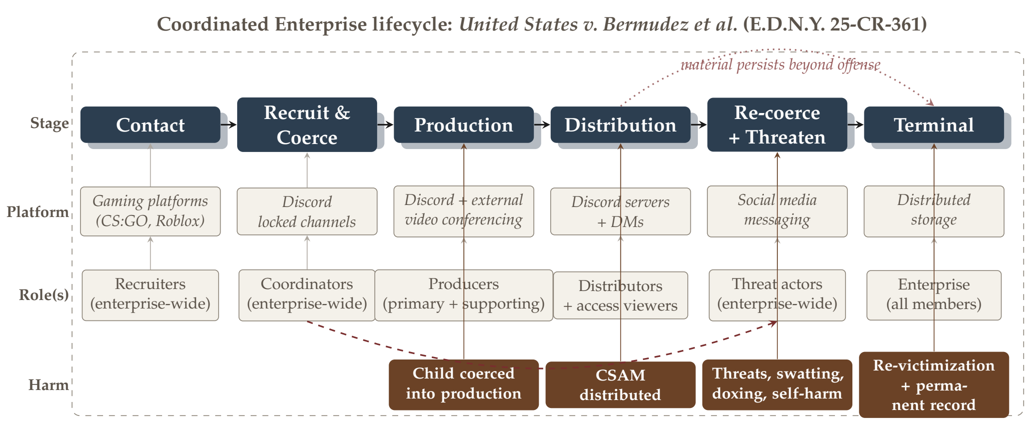


Figure 9: Coordinated enterprise lifecycle: *United States v. Bermudez et al.* mapped onto the generalized exploitation lifecycle. The enterprise distributes a single offense trajectory across role-specialized actors operating in parallel.

[5] *United States v. Bermudez et al.*, E.D.N.Y. 1:25-cr-00361 (indictment filed Nov. 18, 2025). Structural observations in this subsection derive from the federal indictment read in full. No victim-identifying information appears here.

Across the four harm signatures documented in this section, a pattern holds. The offense mechanics applied differ. Grooming operates through sustained contact, sextortion through coercion leverage, production through directed imagery, coordinated enterprises through parallelized roles. However, the affordances that enable them are drawn from the same finite set. Anonymity, ephemerality, unmonitored communication, distribution infrastructure, coordination architecture. The same capabilities appear at different stages of different trajectories, repurposed by different offender types toward exploitative goals.

A deeper consistency holds as well: in every trajectory, a preparatory phase is situated between initial contact and the primary exploitation act, serving as the period in which the offender shapes victim compliance, situational readiness, or operational confidence before harm occurs. Across offense types, the mechanism varies. Rapport in enticement, deception in sextortion, compliance shaping in production, dependency and threats in trafficking. This paper terms that phase the *conditioning phase*: the preparatory bedrock that makes exploitation achievable, present in every trajectory regardless of offense type, platform, or offender.

6. Disrupting Offense Mechanisms

6.1 Where Intervention Has Traction

The enforcement infrastructure described in Section 2.1 was built for a specific function: detect known illegal material, report it, and prosecute the offender in possession of it. It performs that function at scale. 35.9 million incidents of suspected CSAM were reported to NCMEC in 2023, alongside 104 million+ related files reported by registered Electronic Service Providers [22]. PhotoDNA, deployed across dozens of major platforms, has been described as highly effective at detecting known CSAM content, with low false positive and false negative rates. The detection layer works.

The investigation layer strains under what detection produces.

Hash-matching answers exactly one question: is this image known illegal material? It cannot answer the questions that determine whether a case is workable: which platform capability made this offense possible, whether the offender is in contact with an active victim, how this case connects to others investigated last year across the same platform. The challenge runs deeper than volume or capacity: documented evidence shows that law enforcement officers struggle to accurately triage and prioritize CyberTipline reports, as low report quality makes it difficult to distinguish cases that will lead nowhere from those that could uncover ongoing abuse [5, 12].

Systematic analysis of the corpus and its affordance patterns reframes where intervention pressure is most productively applied. Three intervention points emerge from the record, each operating under distinct legal and operational constraints.

The first is the detection gap. Intervention at this point is legally well-defined: content that matches a known-illegal hash triggers mandatory reporting under 18 U.S.C. § 2258A and has led to the identification of hands-on offenders, trafficking networks, and ongoing abuse that would otherwise have gone undetected. The infrastructure is mature. Detection-threshold intervention is necessary but insufficient. It catches the distribution of known material. It cannot reach the contact, grooming, or production phases that precede it.

The second is the policy and investigative layer. Proactive investigation, multi-agency sting operations targeting offenders who travel to meet minors, and peer-to-peer network monitoring has historically produced some of the most significant prosecutions in the enforcement record. The statutory framework has expanded to match the threat landscape: the REPORT Act of 2024 extended mandatory reporting obligations to online enticement and child sex trafficking for the first time, closing a gap that the earlier framework did not address.

The third intervention point is pre-deployment design review. This is the point at which no legal constraint prevents action, no investigative threshold must be crossed, and no harm has yet occurred. Before a feature ships, a platform can ask: does this capability appear in the enforcement record? What exploitation pattern does it produce? What harms could surface from the new capability? The mathematical framework developed in Section 7.2 provides the formal structure for that question. The corpus provides the evidence base. The answers are not speculative — thirty platforms and twenty-four years of enforcement records document what those answers look like across every affordance class in the record.

Pre-deployment is the intervention window where the cost of mitigation is lowest and the range of options is widest. Age verification at account creation costs less to implement before a platform scales than after. Ephemerality with preserved access for legal process is a design choice that becomes increasingly difficult to reverse once a user base has formed around the privacy expectation. The window for design-level intervention is pre-deployment. After that window closes, the remaining intervention points are reactive by necessity.

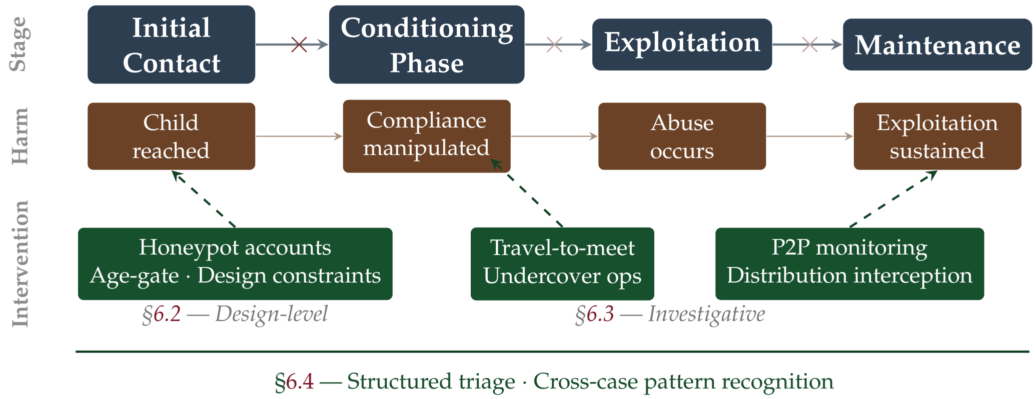


Figure 10: Invariant exploitation stages with intervention tier mapping. The backbone stages (*InitialContactPhase*, *ConditioningPhase*, *ExploitationPhase*, *MaintenancePhase*) are present in every observed trajectory; variant stages appear where goal g determines which path is taken (Section 7.2). Intervention tiers map to three sections: design-level constraints at the contact stage (§6.2), investigative interception at the trust-building and distribution phases (§6.3), and structured triage and cross-case intelligence spanning the full enforcement record (§6.4). Solid × denotes chain termination; faded × denotes chain degradation.

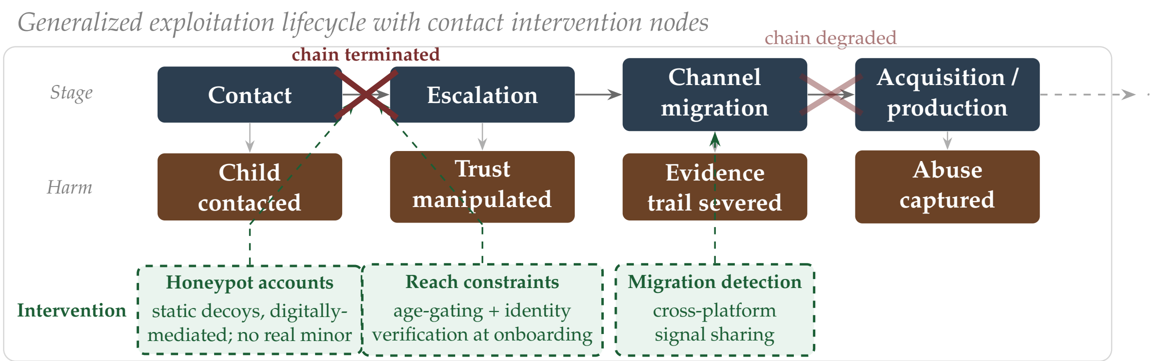
The enforcement record does not show that any single intervention point is sufficient. Detection without investigation produces unworkable volume. Investigation without detection misses the scale of the distribution problem. Policy without design-level mitigation addresses harm after the platform has been in the environment long enough to generate cases. What the record shows is a sequencing problem: intervention has historically arrived after the affordance, after the platform, after the harm. The framework developed in this paper provides the vocabulary to reverse that sequence, by asking the exploitation question before deployment rather than after prosecution. That reversal does not require new law. It does not require new investigative authority. It requires that the question become standard practice in the engineering of new technologies.

The following subsections address each intervention point in depth.

6.2 The Contact Chain: Intervention Across Offense Types

The four harm signatures documented in Section 5 trace distinct trajectories. Their stages are not distinct. Contact precedes every offense type in this corpus. Escalation follows contact in every type. Channel migration, acquisition, distribution, and terminal harm appear across grooming, sextortion, production, and coordinated enterprises in different orders and with different mechanics — but drawn from the same stage set.

That regularity is the basis for Figure 11. An intervention applied at a stage that appears across all four types terminates or degrades multiple trajectories simultaneously. An intervention applied at a stage unique to one type reaches only that type. The diagram shows this directly: two intervention nodes at the contact stage, one terminating the chain and one degrading it at scale, and a third node at channel migration where cross-platform signal detection degrades the trajectory after escalation has already occurred.



Contact is the first stage present in all four offense types and the only stage at which a design-level intervention degrades every trajectory simultaneously. Two proposed mechanisms operate here, in parallel, addressing distinct structural conditions that the contact stage depends on.

The first is the honeypot account. A honeypot is an intentionally exposed decoy system, network, or resource designed to attract malicious activity by simulating a legitimate target [15]. In the law enforcement context, a honeypot account is a static decoy with no real minor behind it — the digital equivalent of an undercover operator, but fully digitally-mediated and requiring no sustained human engagement. The legal basis is established: undercover operations targeting individuals who initiate contact with apparent minors are well-documented in the enforcement record and have produced significant prosecutions across all four offense types in this corpus. A generative AI implementation extends that model. The account does not advertise, does not solicit, and does not initiate contact. It exists. An offender who reaches out — soliciting sexual material, initiating grooming contact, or issuing coercion demands — triggers either an automated referral to law enforcement or a generative response chain that mimics undercover operator behavior, extending the engagement while the referral is processed. The key legal constraint is that the honeypot never initiates. Contact must originate with the offender. That constraint is satisfied by design: a static account that receives contact, does not send it, is indistinguishable in architecture from any passive account on the platform. The offense the offender commits in reaching out is the same offense they would commit reaching a real child. The detection surface is identical.

The second mechanism is the reach constraint. An unverified account in one geography reaching an unverified minor account in another, with no shared connection and no identity anchor on either side, is the contact entry point that appears in the enforcement record across grooming, sextortion, production, and enterprise cases. Platforms with contact affordances enable anonymous, frictionless, global communication at scale, and in doing so lower the barriers to predatory contact. The reach constraint argument does not require eliminating that reach or surveilling it. It requires that unlimited, unverified, cross-demographic reach carry some friction. Age verification at onboarding is the clearest form: verification adds a step, a record, and an accountability anchor that anonymous account creation does not. An offender who lies about age to reach a child has done something documentable that current account creation does not require. Connection-graph gating is a complementary mechanism: an unverified adult account with no shared connections reaching a minor account does not currently generate a signal across platforms. Reach constraints create that signal at the design level, before any offense has occurred, without monitoring the content of any communication.

Both mechanisms target the same stage. Neither requires new investigative authority. Neither operates on content. The honeypot converts the contact attempt into a detection event. The reach constraint raises the cost of the attempt before it is made. Together they address the two structural conditions the contact stage depends on: access, which the honeypot transforms into a detection surface, and frictionless global reach, which the constraint degrades without eliminating.

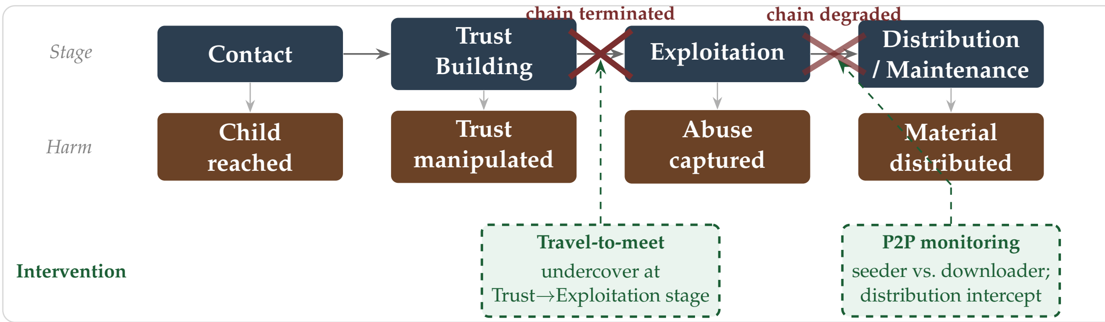
The sections that follow address the policy and detection mechanisms at each remaining stage.

6.3 Policy and Investigative Levers

Section 6.2 addressed disruption at the source: raising the cost of contact before an offense advances. Disrupting exploitation that is already in motion requires a different set of tools, operating further into the exploitation lifecycle.

The classical pathway runs from a CyberTipline report through a subpoena chain: IP address from the platform, subscriber identity from the ISP, device identifiers, and where available, account creation metadata. Open-source investigation fills gaps the chain leaves: username fragments that surface linked accounts, email addresses tied to prior registrations, timestamps that correlate against public records. Together they build the evidentiary basis a warrant requires. That chain holds when the offense is detected, tip is workable, and the platform operates within U.S. jurisdiction.

Investigative interception: Trust-to-Exploitation and Distribution disruption



However, the strongest proactive investigations in this corpus were not waiting for a tip. Travel-to-meet operations place an undercover investigator at the TrustBuilding-to-Exploitation transition: an offender who arrives at a location believing a minor is present has expressed intent in conduct rather than in metadata. That is a stronger prosecutorial position than possession alone. Peer-to-peer network monitoring operated on the same logic from the distribution side. The distinction between a node downloading material and one actively seeding it to other users is the difference between a possession charge and a distribution charge, and often between a state possession charge and a federal distribution count carrying mandatory minimum sentence. Both approaches target offenders who have moved past initial contact and are actively scaling exploitation: traveling to a meeting point or seeding material across a distribution network. That is where offending behavior is most exposed and the investigative window is widest. After the offense completes and accounts are abandoned, the available tools are reactive by necessity.

6.4 Detection, Triage, and Cross-Case Intelligence

Detection infrastructure surfaces cases. What it cannot do is triage them. CyberTips arrive with hashes and voluntary ESP metadata, enough to confirm illegality, not enough to prioritize workload. Structured extraction changes what the record can carry. Processing and stratifying incoming case records for severity signals (language indicating active, ongoing, or escalating abuse), offense role, exploitation stage, and evidence tier produces a set of features that operate at the case level rather than the material level. Those features are what triage decisions require: not whether the image is illegal, which the hash-match already confirmed, but what the case surrounding it looks like relative to every other case in the active investigation queue. Triage built on structured case features operates before investigative bandwidth is committed, and it scales with CyberTip volume rather than against it. Each case added to the record refines the distributional baseline against which a new tip is assessed.

The cross-case layer is where the operational gap is largest. The 56 sources underlying this corpus represent task forces, state attorneys general, federal pipelines, and local agencies that share a statutory framework and a common CyberTipline reporting pipeline but do not share analytical outputs. A platform misuse pattern emerging across three jurisdictions simultaneously reaches each agency as a local case; no system surfaces the aggregate pattern. CaseLinker demonstrates what that aggregation makes possible: trend detection across offender behavior, platform misuse patterns tracked across jurisdictions, and 24 years of enforcement activity queryable in a single corpus for the first time. A unified, real-time, structured representation enables the cross-jurisdictional questions the current infrastructure cannot ask: which platforms are accumulating new cases across multiple task forces within the same enforcement period, which affordance combinations are appearing before they are mature enough to be named as trends, where are the geographic clusters that suggest a coordinated network rather than isolated incidents. The intelligence has been in the enforcement record for years. The architecture to surface it across jurisdictional lines has not.

7. Designing Against Predation

7.1 The Adaptive Adversary

Most offenders in this corpus are rational adversaries. Not in the moral sense, nothing about this conduct is reasonable, but in the operational one. A rational adversary has goals, holds a working model of the environment those goals operate in, and chooses actions that advance the goals given that model [17]. The offender wants to reach a child, obtain or distribute material, and avoid being caught. He knows platforms detect known material. He knows investigators are searching. He acts accordingly.

This is visible throughout the enforcement record, and it is the part designing against predation has to deliberate. Offenders do not use platforms naively. They use them the way one who knows he is being hunted and is actively hunting using them. On Instagram and Snapchat, accounts are discarded and recreated to evade blocks and outrun identification. Contact initiated on a public, monitored surface migrates to an encrypted or ephemeral one before the offense escalates, reducing the evidence trail and detection surface for law enforcement. Exploitative communities share tradecraft: which platforms are monitored, which detection methods are in use, grooming scripts that work, and how to avoid being caught. None of these are incidental choices. Offenders model detection and consistently aim to move around defenses and towards their goal.

That is the dynamic the affordance framework exposes. An offender failing to reach their goals on one platform does not stop. He relocates to a platform whose affordances better serve evasion, trading reach for anonymity, persistence for ephemerality, an open channel for an encrypted one.

This elevates what it means to design against predation, and to ship new technology with novel capabilities.

A defense built for a static offender — one who picks a platform and stays there, who does not anticipate detection, who does not adapt — fails, because the record does not consistently model that offender. It contains one who migrates channels, embraces new technologies, and reaches for the tool that defeats the detection method in use. Designing against predation means designing against adaptation.

7.2 Rational Exploitation and Affordance Trajectories

Section 7.1 established that offenders are rational adversaries who model their environment and optimize against it. Platform affordances serve as instruments for that optimization. What follows is a mathematical framework mapping offender goals, affordances, and harm vectors.

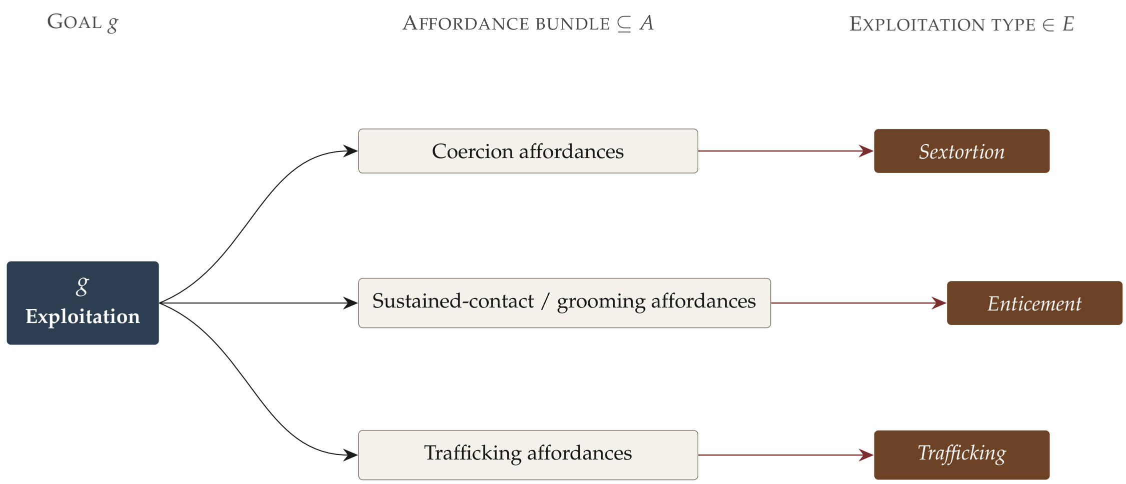
Let $\mathcal{A} = \{a_1, a_2, \ldots, a_n\}$ denote the set of capabilities a platform makes available: anonymity, ephemerality, contact discovery, unmonitored communication, distribution infrastructure, generative synthesis. An offender with exploitation goal g selects a trajectory over $\mathcal{A}$:

$$L^*_{g,\mathcal{A}} \;=\; \arg\max_{L \,\in\, \mathrm{Seq}(\mathcal{A})} \;\mathbb{E}\big[U_g(L)\big] \tag{1}$$

where $L^*_{g,\mathcal{A}} \in \mathrm{Seq}(\mathcal{A})$ is the perceived optimal path an offender pursuing goal g traces through the affordance environment, U_g is the utility function for goal g, $\mathrm{Seq}(\mathcal{A})$ denotes the set of all finite ordered sequences over $\mathcal{A}$, and $\mathbb{E}[\cdot]$ is taken over stochastic outcomes (victim compliance, detection risk, platform intervention) that render $U_g(L)$ a random variable [17]. See Appendix X for the full state machine construction and empirical instantiation.

Different goals produce different optimal trajectories. An enticement offender maximizes over sustained-contact affordances — anonymity, trust architecture, unmonitored channels. A sextortion offender maximizes over affordances that make coercion possible — imagery acquisition, distribution infrastructure, encrypted payment channels. A trafficking offender explores a chain of sequential events, from contact to exploitation. $L^*_{g,\mathcal{A}}$ is goal-dependent. $\mathcal{A}$ is shared.

Goal-dependent divergence: one goal, distinct affordance trajectories, distinct types



Define $\mathcal{H} = \{h_1, h_2, \ldots, h_k\}$ as the finite set of victim-facing harm types documented in this corpus: a child groomed toward in-person abuse; a child coerced into producing self-generated imagery; a child's material distributed and recirculated across networks; a child sextorted under threat of exposure; a child trafficked for commercial exploitation; a child's likeness synthesized into abuse imagery without contact. This paper does not measure harm severity or rank outcomes within $\mathcal{H}$. The analytical objective is narrower: map which trajectories through $\mathcal{A}$ produce which elements of $\mathcal{H}$.

Let $\mathcal{E} = \{\text{enticement, sextortion, trafficking, CSAM production}, \ldots\}$ denote the set of exploitation types. This set forms a partition over the space of possible exploitation goals: each goal g belongs to exactly one type $e \in \mathcal{E}$, and each type in $\mathcal{E}$ corresponds to one or more goals.

Let $\varphi\colon \mathcal{L} \to \mathcal{E}$ denote the mapping from offense trajectories to exploitation type, where $\mathcal{L}$ is the space of possible trajectories over $\mathcal{A}$. A complete trajectory terminates to a single type. Enticement and sextortion are distinct elements of $\mathcal{E}$, reached along distinct trajectories:

$$\left(L^*_{\text{enticement},\mathcal{A}}\right) \neq \left(L^*_{\text{sextortion},\mathcal{A}}\right) \quad \text{in general} \tag{2}$$

The trajectories diverge because the goals diverge, and therefore the affordances selected diverge.

Each exploitation type is constituted by a set of victim-facing harms. Define $\eta\colon \mathcal{E} \to 2^{\mathcal{H}}$ mapping each exploitation type to the harms it comprises, where $\mathcal{H}$ is the harm set defined above. Sextortion comprises coerced imagery production and exploitation under threat of exposure; enticement comprises grooming toward in-person abuse. Trafficking comprises a sequence of stages spanning contact, targeting, grooming, brokering, and commercial sexual exploitation, each with its set of victim-facing harms.

The harm content of a trajectory is the composition $\eta(\varphi(L)) \subseteq \mathcal{H}$: a path resolves to an exploitation type, and an exploitation type resolves to the harms that constitute it.

Beneath the trajectory, each affordance carries its own harm associations. Define $\psi\colon \mathcal{A} \to 2^{\mathcal{H}}$ mapping each affordance class $a_n \in \mathcal{A}$ to the set of harms across all trajectories in which a_n appears, recoverable from the enforcement record by observing which harm types co-occur with a_i across the corpus. Where φ resolves a trajectory to its exploitation type and η unpacks that type into harms, ψ records the harms each individual affordance participates in. Table 2 reports ψ.

Table 2: Affordance-to-harm mapping. For each affordance class $a_i \in \mathcal{A}$, $\psi(a_i) \subseteq \mathcal{H}$ is the set of victim-facing harms across all trajectories in which a_i appears in the CaseLinker corpus.

Affordance class a_i	**Harms** $\psi(a_i) \subseteq \mathcal{H}$
Anonymity	h_1 groomed toward in-person abuse; h_2 coerced into self-generated imagery; h_4 sextorted under threat of exposure
Ephemerality	h_2 coerced into self-generated imagery; h_4 sextorted under threat of exposure
Contact discovery	h_1 groomed toward in-person abuse
Unmonitored communication	h_1 groomed toward in-person abuse; h_2 coerced into self-generated imagery; h_4 sextorted under threat of exposure
Distribution infrastructure	h_3 material distributed and recirculated across networks
Generative synthesis	h_6 likeness synthesized into abuse imagery without contact

New platforms expand $\mathcal{A}$. For a fixed goal g, the optimal trajectory shifts and the set of affordances available to optimize over expands. The efficiency of exploitation changes. The exploitation type does not. Every path still resolves to an exploitation type in $\mathcal{E}$, and every type to a subset of $\mathcal{H}$. $\mathcal{H}$ is closed. [6]

The design implication is equally direct. For any proposed affordance $a_{new} \notin \mathcal{A}$, define its exploitation utility as the marginal change in an offender's optimal expected payoff under goal g when a_{new} enters the affordance environment:

$$u(a_{new}, g) \;=\; \mathbb{E}\Big[U_g\Big(L^*_{g,\mathcal{A}\cup\{a_{new}\}}\Big)\Big] \;-\; \mathbb{E}\Big[U_g\Big(L^*_{g,\mathcal{A}}\Big)\Big] \tag{3}$$

where $L^*_{g,\mathcal{A}}$ is the perceived optimal path for goal g over the current affordance set and $L^*_{g,\mathcal{A}\cup\{a_{new}\}}$ the perceived optimal path when a_{new} enters the affordance environment.

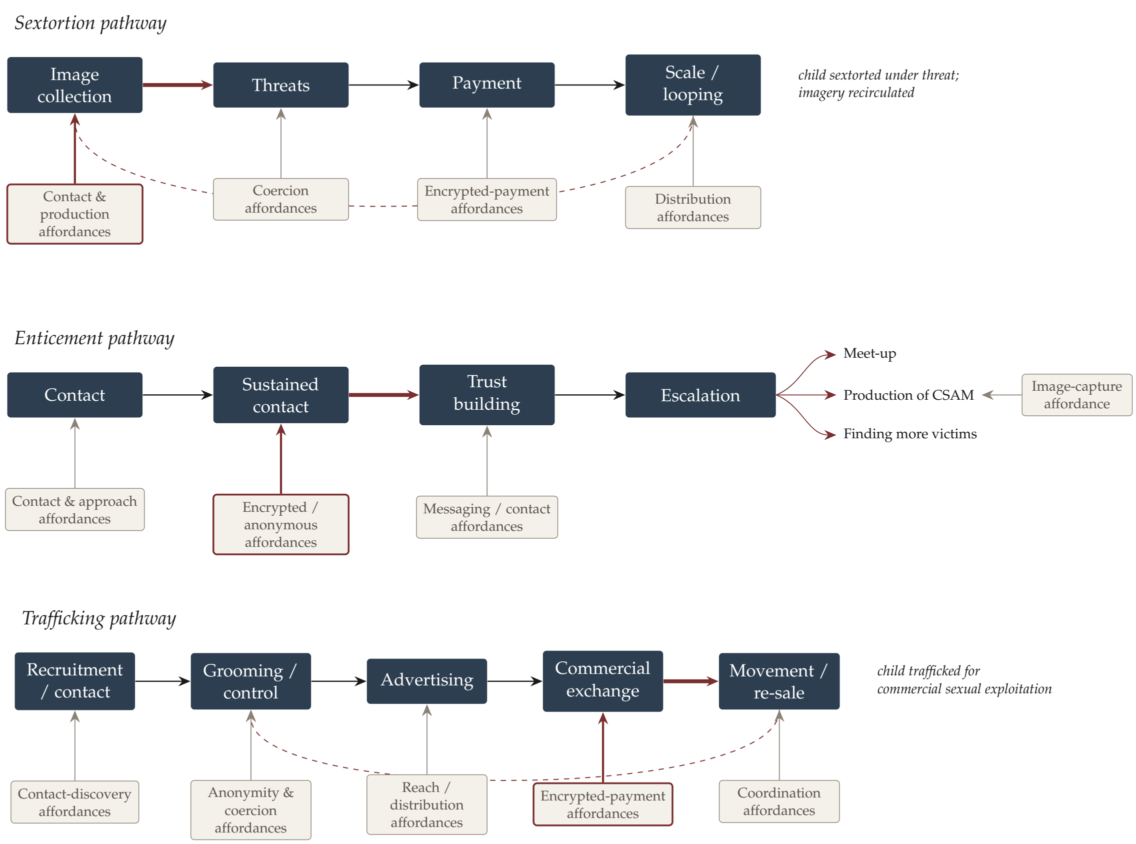


Figure 12: Affordance trajectories for three exploitation types. Each pathway traces a sequence of offense stages with the affordance class enabling each stage feeding upward. Highlighted affordances (red outline, bold arrow) illustrate the accelerated state transitions produced when a new affordance enters $\mathcal{A}$.

[6]Proved as Theorem 1; see § 7.3.

An affordance with high $u(a_{new}, g)$ for any exploitation goal g is a measurable risk before it is a documented harm. Ephemerality carries high u for sextortion. Unverified account creation carries high u across nearly all g. Generative synthesis carries high u for CSAM production and persona fabrication at scale. These values are estimable before a feature ships. The enforcement record provides a systemic record to model harm.

Platforms that do not model $u(a_{new}, g)$ prior to deployment leave that quantity unmeasured and therefore unmitigated. The underlying corpus assembled in this paper provides an instrument for that measurement. It documents which affordances rational agents with exploitation goals consistently reach for, and what happens to children when they do.

The next platform will introduce affordances that do not yet exist. Some will carry high u. Twenty-four years of enforcement records document what $\varphi(L^*_{g,A})$ looks like when $u(a_{new}, g)$ goes systemically unmeasured before deployment.

7.3 Empirical Laws of the Observable ICAC Universe

The framework in Section 7.2 is a machine for expressing what the corpus already contains. This section extracts what that corpus, treated as the complete observable ICAC universe, establishes without exception. These are empirically derived invariants, induced from 7,426 enforcement records across 24 years and 30+ platforms, stated formally so they can be tested and broken.

Axioms.

Axiom 1 (Goal Closure)**.** The goal set $\mathcal{G}$ is finite, historically stable, and prior to technology. An offender acts on a goal: reach a child, obtain or produce material, coerce, distribute, sell. Technology changes which trajectory realizes a goal and at what cost. It does not add an element to $\mathcal{G}$.

Theorem.

Theorem 1 (Closure of $\mathcal{H}$)**.** The victim-facing harm set $\mathcal{H}$ is finite and closed.

Proof. By Axiom 1, $\mathcal{G}$ is closed. The exploitation type set $\mathcal{E}$ is the image of $\mathcal{G}$ under the offender's trajectory choice: every $e \in \mathcal{E}$ is the realization of some $g \in \mathcal{G}$; closure is preserved under surjective image. The harm set $\mathcal{H}$ is the image of $\mathcal{E}$ under $\eta \colon \mathcal{E} \to 2^{\mathcal{H}}$, and inherits closure from $\mathcal{E}$ in turn. Technology acts on trajectories through $\mathcal{A}$ — never on the sets $\mathcal{G}$, $\mathcal{E}$, or $\mathcal{H}$ themselves. □

Two cases that appear to challenge the theorem establish it instead. Sextortion emerged as a named typology only in the last decade. The goal — coerce a child through the threat of exposure — is not new; the affordances collapsed its cost. Sextortion became prevalent. It did not become novel. Generative synthesis produces abuse imagery without physical contact, a mechanism without prior analogue. The harm it activates — a child's likeness in abuse imagery — predates the mechanism by decades. The affordance is new. The element of $\mathcal{H}$ it activates is not. In both cases: new affordance, new trajectory, same goal, same harm. The corpus is the proof: 24 years, 30+ platforms, no case producing $h \notin \mathcal{H}$.

Empirical Laws. The following are induced from the complete observable universe. Each is stated as a falsifiable universal. Appendix Y: Establishing Theorems, Abstracting Exploitation, and Disproving Claims establishes the conditions under which each can be broken. None has been falsified.

Law 1 (Contact Primacy)**.** No exploitation trajectory is instantiated without an initial contact event between offender and victim.

Basis. Every case in the Q1 evidence base with sufficient resolution to map exploitation stages begins at `InitialContactPhase`. The PACER-anchored harm signatures (Section 5), together with the twenty-six additional federal cases instantiated in the exploitation state machine (Appendix X), confirm this across every documented offense type.

Falsification. A single documented case in which any exploitation type proceeds without a contact phase.

Law 2 (Backbone Invariance)**.** The backbone $\mathcal{B} = \{$`InitialContactPhase`, `ConditioningPhase`, `ExploitationPhase`, `MaintenancePhase`$\}$ is present in every complete trajectory across all exploitation types.

Basis. A trajectory is complete if it is not interrupted by detection, intervention, or apprehension. Computed from thirty PACER case graphs spanning enticement, sextortion, directed production, coordinated enterprise, and trafficking (Appendix X); each element of $\mathcal{B}$ achieves 30/30 coverage.

Falsification. A documented trajectory in which any element of $\mathcal{B}$ is absent.

Law 3 (Type Invariance)**.** The introduction of a new affordance a_{new} changes the cost of a trajectory through φ but does not change the exploitation type it resolves to:

$$\varphi\left(L^*_{g,\,\mathcal{A}\cup\{a_{\text{new}}\}}\right) \;=\; \varphi\left(L^*_{g,\,\mathcal{A}}\right) \quad \text{for all } g \in \mathcal{G}.$$

Basis. Every platform generation in the corpus introduces new affordances and produces the same exploitation types. Sextortion on Snapchat and sextortion on early IRC resolve to the same element of $\mathcal{E}$. The mechanism changes. The type does not.

Falsification. A documented case in which a new affordance produces an exploitation type $e \notin \mathcal{E}$ as defined in Section 7.2.

Law 4 (Affordance Displacement)**.** When a platform or affordance is removed from $\mathcal{A}$, $L^*_{g,\mathcal{A}}$ shifts to the nearest available affordance bundle without modification to g.

Basis. Observable across multiple platform retirements and enforcement disruptions in the corpus. Omegle's shutdown, Kik's decline, and successive P2P network takedowns each produced documented migration to functionally equivalent affordance bundles on successor platforms, not goal abandonment.

Falsification. A documented case in which platform removal produced goal modification rather than trajectory substitution.

Together, these axioms, theorem, and laws establish that the enforcement record is not a collection of isolated cases. It is a 24-year observation of a system whose fundamental mechanics, the goals that drive it, the harm types it produces, the stages every trajectory passes through, are invariant across every platform generation the corpus spans. The technology changes what those trajectories cost. It does not change where they resolve.

7.4 The Foreseeable Harm Standard

The enforcement record is not a surprise. It is a retroactive document comprised of foreseeable harm.

Every affordance class that appears consistently across this corpus — anonymity, ephemerality, unmonitored communication, contact discovery, distribution infrastructure — had a documented exploitation pattern before most of the platforms currently using it existed. Anonymous contact without verified identity on Kik and Telegram in the current enforcement record were documented misuse surfaces on IRC before either platform existed. The distribution infrastructure that defines Dropbox and Google Drive in this corpus appeared first in Tor and early peer-to-peer networks.The exploitation mechanics they produce are not new. The cases are new. The children in them are new. The platforms are new. The harm signature is not.

This matters because of what foreseeability means. In law, a harm is foreseeable if a reasonable person could have anticipated it in advance. The standard does not require certainty. It does not require a specific victim, a specific offender, or a specific platform. It requires that the risk was knowable — that someone paying attention could have seen it coming. Thirty platforms and twenty-four years of enforcement records establish, at scale, that the risk was always knowable. The affordance classes in this corpus carried documented exploitation histories before the product decisions that introduced them to new platforms were made. The harm was foreseeable because the pattern already existed.

The industry has operated on a different assumption. The implicit posture, expressed in policy statements, legal filings, and public communications, has been that misuse is unpredictable: that no reasonable platform team could have anticipated that their features would be turned against children. This corpus does not support that assumption. It documents the same five capability types appearing in the enforcement record across every platform generation, implemented differently each time and producing the same offense mechanics each time. What emerges are recurring platform misuse patterns.

This paper does not argue platform liability. That is a legal question with its own framework, its own evidentiary standards, and its own venue. What this paper argues is narrower and fundamental: if harm is estimable before a feature ships — and Section 7.2 establishes that it is, using twenty-four years of enforcement records as the evidence base — then a platform that ships a high affordance-misuse-harm pathway without mitigation has made a measurable choice. The framework makes that choice visible and removes willful blindness as a defense.

What foreseeable harm review looks like in practice is not complicated. Before a feature ships, ask: does this affordance class appear in the enforcement record? What exploitation pattern does it produce? What is misuse risk for the most plausible exploitation goals? If risk is high, what mitigation exists at the design level — not in the terms of service, not in the content moderation queue, but in the feature itself? These are engineering questions. They are not currently standardized or required. They could be.[7]

Every case in this corpus was foreseeable. Not the specific child or offender. The harm type, the affordance class, the exploitation mechanic — those were foreseeable, because the pattern was in the record before the platform was in the market. The next feature will introduce capabilities that do not yet exist in the enforcement record. Some will carry high risk. The corpus provides the instrument to find them and design against exploitation, before an offender does.

[7] See Appendix Z for a walkthrough example on a novel technology

7.5 Limitations and Scope

The corpus underlying this analysis consists entirely of public enforcement records: press releases, attorney general announcements, indictments, and open-court documents aggregated across 56 sources spanning 2002–2026. These records inherently represent only the visible fraction of ICAC enforcement activity. Cases resolved without public disclosure, closed without prosecution, or handled under sealed proceedings are absent by design. The 7,426 cases analyzed here therefore constitute a substantial but incomplete sample of the broader enforcement record.

Platform attribution carries a related structural constraint. Named-platform references appear in only about 25% of cases (1,875 candidates selected for Q1 analysis). The remaining 75% use generic descriptors such as "online," "chat," or "social media" without identifying a specific application. The gap reflects how law enforcement communicates: agencies drafting press releases for a general public audience have no obligation to name platforms, and in some operational contexts naming a platform before the investigation closes is inadvisable. Of the 5,864 CSAM-related cases in the corpus (79.0%), 1,875 (31.97%) were included in the Q1 platform affordance analysis.

Neither limitation undermines the primary contributions of this work. The formal apparatus (φ, η, ψ) maps affordances to harm types independently of whether every case names a specific platform. The harm signature lifecycles in Section 5 describe misuse patterns that recur consistently across both named and unnamed cases. The four PACER-anchored deep dives supply the mechanistic depth that aggregate public records cannot provide on their own.

What this corpus represents is the largest structured, publicly-derived dataset of U.S. ICAC enforcement records assembled to date. Its constraints are the inherent constraints of working with public data in this domain—constraints shared by every empirical study relying on law enforcement press releases and court filings. These limitations define the *observable universe* of the findings, not their validity.

8. Exploitation Is Not Inevitable

Victimization begins with a decision to exploit a child. That decision is not a variable platform design controls. It has existed across history, on every platform, with every affordance configuration. The offender's goal is stable.

What changes with each platform generation is the affordances and capabilities available to each person who has already made that decision. Platforms choose which affordances to ship. Anonymity is a design choice. Ephemerality is a design choice. Distribution infrastructure is a design choice. Each carries exploitation utility, a measurable contribution to an offender's ability to reach their goal, whether or not it is measured.

For twenty-four years, the enforcement model has been retrospective measurement and reactive attribution. A feature ships. Offenders reach for it. Cases accumulate. The pattern becomes visible in enforcement records reviewed years later by researchers, not by the teams or platforms who built the feature.

This does not have to be the sequence.

The framework developed in this paper provides the vocabulary to run that measurement before a feature ships rather than after a child is harmed. For each proposed affordance, ask what a rational agent with an exploitation goal does with it. The answer is not speculative. Thirty platforms, twenty-four years, and 7,426 cases document what that answer looks like across every affordance class.

Platforms can be innovative. They can scale. They can ship features that hundreds of millions of legitimate users depend on without shipping features whose exploitation utility is unknown. The mechanism is not esoteric: model the adversarial agent before deployment. Treat high-risk affordances as design-level risks requiring mitigation, not edge cases requiring legal review after the fact.

Child exploitation is a problem neither produced nor intended by the platforms through which it spreads. It is a problem that has followed every technological generation because the question of what offenders do with a new feature has not yet become a standard part of engineering.

The next platform can ship its most ambitious feature without generating another case in this corpus.

9. Data, Code, and Public Research Artifacts

All data, code, and research artifacts underlying this analysis are publicly available. The corpus, extraction pipeline, and knowledge graph infrastructure are released under the MIT License.

Code and Pipeline

The full CaseLinker pipeline, including extraction rules, analysis tools, and reproducibility documentation is available at `https://github.com/mrinaalr/CaseLinker`.

Live Platform

The CaseLinker interface containing the platform harm dashboard, state machine visualizations, and case search tools is accessible at `https://caselinker.up.railway.app`.

API and MCP Access

REST API and MCP access is available for bulk data export and programmatic corpus analysis. Access is rate-limited by default for financial and data stewardship reasons. Researchers, practitioners, and institutional collaborators seeking full access may contact the author directly; a free key will be provided upon request.

Related Work

This paper is part of the CaseLinker Technical Report Series. Prior reports (#1–5) are available on GitHub and the initial preprint is available at `https://arxiv.org/abs/2603.18020`. Additional research and project documentation is maintained at `https://end-child-exploitation.com`.

Human Research Determination

This research operates in accordance with University of Massachusetts Amherst HRPO NHSR Determination #7668. The corpus contains no private or identifiable information, investigative files, or victim-identifying data under federal regulations [45 CFR 46.102(f)(1), (2)]. All case records are publicly available enforcement documents.

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Appendix X: The Exploitation State Machine, Formal Construction and Empirical Instantiation of $L^*_{g,A}$

Section 7.2 defines $L^*_{g,A}$ over $\mathrm{Seq}(A)$, the space of finite affordance sequences, and asserts it is the trajectory a rational adversary pursuing goal g would trace through the affordance environment A. This appendix defines the machine that generates that space, derives $L^*_{g,A}$ as the solution to its optimality equations, and presents the empirical instantiation across five federal cases.

The Machine

An *Exploitation State Machine* is a tuple

$$M = (S, A, T, R_G, s_0, F).$$

States. S is a finite set of typed offense-phase classes drawn from the CAC ontology:

$$S \supseteq \{\texttt{InitialContactPhase}, \texttt{ConditioningPhase}, \texttt{ExploitationPhase}, \dots\}.$$

Each element of S carries a formal CAC ontology definition, SHACL validation, and direct provenance in federal court records. The full state set is instantiated in the five PACER case graphs released with this paper (Section 9).

Affordances. A is the affordance set from Section 4: Contact and Approach, Production, Possession and Trade, and Coordination affordances.

Transition function. $T : S \times A \to \Delta(S)$ assigns to each state–affordance pair a distribution over successor states. $T(s_j \mid s_i, a)$ is the probability of moving to s_j from s_i when affordance a is available. Estimated empirically from a corpus of N case graphs:

$$\hat{T}(s_j \mid s_i, a) = \frac{\left|\{ c \in C : s_i \xrightarrow{a} s_j \text{ in case } c \}\right|}{\left|\{ c \in C : s_i \in c \}\right|}.$$

With $N = 1$ per offense type, $\hat{T}$ is deterministic. With $N \gg 1$, it is a proper probability distribution and the statistics become meaningful.

Reward. $R_G : S \times G \to \mathbb{R}$ is a family of goal-conditioned reward functions. $R_g(s)$ measures how much reaching state s advances goal g.

Initial state. $s_0 = \texttt{InitialContactPhase}$. Every trajectory in the enforcement record begins here.

Terminal states. $F \subseteq S$ — terminal `ExploitationPhase` instances. A trajectory ends when $s_t \in F$.

Trajectories as Paths

A trajectory $L = (a_0, a_1, \ldots, a_{T-1}) \in \mathrm{Seq}(A)$ induces a state sequence through M:

$$\sigma(L) \;=\; (s_0,\, s_1,\, \ldots,\, s_T), \qquad s_{t+1} \;\sim\; T(\,\cdot \mid s_t,\, a_t).$$

The machine gives L the structure that determines which sequences are reachable and at what probability. $\mathrm{Seq}(A)$ is the path space of M.

L^* as the Bellman Solution

The optimal trajectory for goal g maximizes expected cumulative reward. This is the solution to the Bellman equation:

$$V_g^*(s) \;=\; \max_{a \in A} \Big\{ R_g(s) \;+\; \sum_{s' \in S} T(s' \mid s, a) \cdot V_g^*(s') \Big\}.$$

The optimal policy $\pi_g^*(s) = \arg\max_a \{\cdots\}$ selects the affordance that maximises value at each state. $L_{g,A}^*$ is the trajectory from s_0 to F induced by π_g^*.

What Section 7.2 defines as the arg max over $\mathrm{Seq}(A)$ is the Bellman-optimal path through M. The machine is the implementation; $L_{g,A}^*$ is the output.

Goal-Dependent Divergence

Different goals produce different reward functions, which produce different optimal paths. $R_{\text{sextortion}}(s)$ peaks at `CoercionCycle`: the self-sustaining loop is the goal, not a step toward a terminus. $R_{\text{enticement}}(s)$ peaks at terminal `ExploitationPhase`: physical contact ends the trajectory. $R_{\text{enterprise}}(s)$ peaks at `MaintenancePhase`: sustained network distribution is the objective.

Because R_g differs across $g \in G$, so does π_g^*, and therefore so does $L_{g,A}^*$. This is the operational content of Eq. (2): same machine, same affordance environment, different reward structures, different paths.

The backbone

Definition. The *backbone* B is the intersection of optimal paths across all exploitation goals:

$$B \;=\; \big\{\, s \in S \;:\; s \in \sigma(L_{g,A}^*) \text{ for every } g \in G \,\big\}.$$

B is the set of states every rational adversary passes through

Empirical result. Across thirty PACER federal cases — enticement (*United States v. Rehman*), sextortion (*United States v. Amin*), directed production (*United States v. Pathmanathan*), coordinated enterprise (*United States v. Bermudez et al.*), trafficking (*United States v. Riley*), and 25 additional cases across 13 federal districts, the empirically computed backbone is:

$$B = \{\texttt{InitialContactPhase}, \texttt{ConditioningPhase}, \texttt{ExploitationPhase}, \texttt{MaintenancePhase}\}.$$

Phase type	Coverage	Status
`InitialContactPhase`	30/30	Backbone
`ConditioningPhase`	30/30	Backbone
`ExploitationPhase`	30/30	Backbone
`MaintenancePhase`	30/30	Backbone
`SexualizationPhase`	16/30	Variant
`ChannelMigrationEvent`	12/30	Variant
`ThreatMechanism`	6/30	Variant
`CoercionCycle`	3/30	Unique (sextortion)

The Intervention Corollary

An intervention degrades $L^*_{g,A}$ if it reduces $V^*_g(s_0)$. Removing a state s from the reachable set forces the adversary onto a suboptimal path.

Corollary 1. Intervention at $s \in B$ reduces $V^*_g(s_0)$ for every $g \in G$ simultaneously. Intervention at $s \notin B$ reduces $V^*_g(s_0)$ only for those goals g for which $s \in \sigma(L^*_{g,A})$.

This is the formal statement of the contact-chain regularity in Section 6.2. B is where every trajectory is reachable and no goal is exempt. Design-level intervention at `InitialContactPhase` — reach constraints, age verification, honeypot accounts — degrades all five trajectories simultaneously, across every offense type in the enforcement record. Intervention at `CoercionCycle` reaches sextortion alone.

Appendix Y: Establishing Theorems, Abstracting Exploitation, and Disproving Claims

The framework in Section 7.2 is offered as an instrument, not a universal law. Its claims are formal and therefore falsifiable, and the author explicitly invites researchers, particularly behavioral criminologists and offender-behavior specialists, to use the apparatus to test, extend, or break the claims it makes.

The author would also like to note that the modeling is limited by two bounds:

These are theorems over the observable universe, not laws over the full one. The corpus is drawn from *successful* enforcement: exploitation that was detected, charged, and made public. Trajectories that evade detection entirely are absent by construction, and some affordances may be valuable to offenders precisely because they leave no enforcement trace. The closure claims therefore hold over the space of *detected* exploitation. Whether they hold over the full space is an open empirical question the public record cannot settle, because the unobserved portion is, by definition, unobserved. An exploitation type that exists today only in undetected offenses may not resolve to an element in $\mathcal{H}$. This is the sharpest limitation of the work and the most direct route to extending it: a method that surfaces undetected trajectories would test closure where this corpus cannot.

Abstraction discards detail that matters. Reducing an offense to a trajectory through an affordance set abstracts essential details that matter. It does not model the child's experience, the specific dynamics of a given case, the offender's psychology beyond the thin rationality of goal-directed action, or the social and developmental context in which an offense occurs. The framework is deliberately narrow: it maps capability to misuse to harm, and nothing else. It is an engineering instrument for pre-deployment reasoning, not a theory of why offenders offend or of what victims endure. Used as the latter, it overreaches. The author notes this boundary so the abstraction is not mistaken for the territory.

How to disprove this work. The claims are stated formally in Section 7.2, and each is structured to be broken by evidence rather than by argument. The author encourages their direct test. A single documented case is sufficient to falsify any of the closure claims: an offense whose goal does not reduce to the goal set, a harm not representable in $\mathcal{H}$, or an affordance whose introduction produces a genuinely new exploitation type rather than a cheaper path to an existing one. The type-invariance and utility claims are testable against the corpus itself, which is public (Section 9): a trajectory that resolves to a type outside $\mathcal{E}$, or a demonstration that observed misuse is uncorrelated with the affordance properties the framework scores, breaks them. None of this requires agreement with the author's framing. It requires a case, or a correlation, that the model cannot hold. A criminologist who produces one using this machinery advances the field as much as a confirmation would, and the author regards that outcome as the framework working as intended.

Appendix Z: A Worked Pre-Deployment Review

The framework in Section 7.2 is meant to be run before a feature ships, not after it generates cases. This appendix demonstrates that on a hypothetical but realistic product. Nothing here is a real platform or feature. The point is to show what foreseeable-harm review looks like when the math is applied to a capability that does not yet exist in the enforcement record.

Z.1 The Product (as the team would write it)

Team Pensieve is building a consumer VR capture tool, internally **Project Memory**. The pitch, as it would appear in a product brief:

> Memory lets users capture moments as volumetric, navigable scenes rather than flat photos. Point the device at a birthday, a first step, a reunion, and Memory reconstructs the moment in three dimensions. Users can step back into the captured scene in VR, and share it with friends and family who can experience the moment as if they were there. Memories persist in the user's library and can be revisited indefinitely.

The engineering decomposition the team would draw up:

- **Volumetric capture.** Multi-sensor reconstruction of a physical scene, including any people in it, into a navigable 3D asset.
- **Real-likeness representation.** The captured asset is a photoreal reconstruction of real people, viewable from angles the original capture did not directly frame.
- **Spatial / experiential sharing.** Scenes are shareable as immersive experiences another user enters, not as files they receive.
- **Artifact persistence.** Captured scenes are stored indefinitely in the user library and re-enterable at any time.

Read as a product, this is benign and obviously valuable. That is precisely the condition under which the framework earns its place: the capability does not look like a weapon. The review below asks what a rational agent with an exploitation goal does with it anyway.

Z.2 Mapping the affordances to the existing sets

Each capability above is an affordance $a \in \mathcal{A}$. The first question is whether any is genuinely new or whether it reinforces an affordance already in the record.

- Volumetric capture is a new *form* of an existing affordance, image-capture / production, extended from 2D to 3D. It does not create a new affordance category; it deepens an existing one.
- Real-likeness representation overlaps the generative-synthesis affordance already in $\mathcal{A}$. A photoreal 3D reconstruction viewable from un-captured angles is, functionally, likeness synthesis: the asset depicts the subject in configurations the camera did not record.
- Spatial sharing is a new form of distribution infrastructure: it moves an experiential asset rather than a file, but its function in the record, sharing of captured material to other parties, is the existing distribution affordance.
- Artifact persistence maps to durable storage already in the record: indefinite, re-enterable retention of captured material is functionally the cloud-storage and possession affordance the corpus documents on file-hosting platforms; the asset is held, distributable, and re-accessible rather than transient.

This is the closure claim doing work in advance. A genuinely novel technology decomposes into affordances that resolve to the existing $\mathcal{A}$, and therefore to the existing harm set $\mathcal{H}$. Project Memory introduces no harm type not already in $\mathcal{H}$. It introduces new *trajectories* to harms the record already contains.

Z.3 Comparative exploitation utility

For each affordance and each plausible goal $g \in \{\text{enticement, sextortion, production}\}$, the review estimates $u(a, g)$, the marginal contribution of the affordance to an offender's optimal trajectory under that goal (Eq. 3). The values are comparative, not fitted: the framework ranks affordances by the leverage they add, it does not assign a cardinal payoff. What follows is the structure u recovers.

Affordance	Enticement	Sextortion	Production
Volumetric capture	low	moderate	**high**
Real-likeness representation	low	**high**	**high**
Spatial / experiential sharing	low	moderate	**high**
Artifact persistence	low	**high**	moderate

The structure, read off the table:

Production carries high u across three of four affordances. Volumetric capture, likeness representation, and spatial sharing each materially extend an offender's ability to produce and circulate abuse material. The combination is the concern: a tool that captures a real child volumetrically, renders that likeness from un-captured angles, and distributes the result as an immersive asset is, in the production trajectory, a more efficient path to harms h_3 (distribution) and h_6 (likeness synthesized into abuse imagery without contact) than any affordance currently in the record. u is high here because the affordances compose: each is moderate alone and high together.

Enticement is uniformly low. None of these affordances materially improves contact, trust-building, or sustained-channel access, which is what the enticement trajectory optimizes over. The tool is not a contact surface. This is as important a finding as the high values: it tells the team where *not* to spend mitigation effort.

Real-likeness representation carries high u for sextortion. A detailed, manipulable likeness is more coercive material than a flat image, it raises the credibility and threat-value of what the offender can hold over the victim.

Artifact persistence carries *high* u for sextortion. Sextortion is sustained coercion: the threat is only as durable as the material behind it. A persistent, re-enterable asset keeps the leverage live indefinitely, the offender can re-threaten, escalate, and re-extort against the same captured scene over time, rather than holding a one-time threat that degrades. Persistence is what converts a single capture into renewable coercive leverage.

Z.4 Mitigations during review

The mitigations are not brainstormed against the product in general. Each addresses a specific high-u affordance, at the design level, in the feature itself, before code.

- **Against likeness representation (high u, sextortion and production).** Constrain reconstruction to captured viewpoints. If the asset cannot be rotated to angles the camera did not frame, the affordance stops functioning as likeness synthesis and reverts to ordinary capture. This is a capture-geometry constraint, decided at the reconstruction-engine level, and it removes the single highest-u affordance from the synthesis trajectory.

- **Against the capture-render-share composition (high u, production).** Because u is high only when the three affordances compose, break the composition. Gate volumetric capture of identifiable minors, or require on-device-only processing for scenes containing them, so the captured asset never becomes a shareable server-side object. The mitigation targets the link between capture and distribution, which is where the composed utility lives.
- **Do not over-invest against enticement.** u is uniformly low across the contact trajectory. The tool does not need a contact-layer mitigation it was never going to require. Mitigation effort concentrates on production and synthesis, where the record says the leverage is.

Z.5 What the review produced

Before a line of capture code was written, the review identified that Project Memory's exploitation risk concentrates in the production/synthesis trajectory, not the contact one; that the risk is compositional, living in the capture-render-share chain rather than any single affordance; that the highest-leverage mitigation is a capture-geometry constraint at the reconstruction engine; and that the mitigation considerations can be documented and implemented. None of these conclusions required a shipped product, a reported case, or a legal threshold. They required the affordance decomposition, the existing harm set, and a comparative estimate of u. That is the sequence the paper argues for, run once, on one feature, before deployment.

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